

Response to Reviewers

Interspecies Interactions Drive Community-Level Selection in Microbial Coalescence

Dear Editor and Reviewers,

We thank the reviewers for their careful, rigorous, and constructive feedback. The comments substantially improved both the framing and the technical clarity of the manuscript. Below, we respond point by point and summarize the analyses and text revisions undertaken in response to the major concerns.

Reviewer comments are shown in blue. Author responses are shown in black. Text quoted from the revised manuscript is shown in dark red, matching the marked revised manuscript.

Summary of major revisions.

- We clarified the central conceptual framing by defining community-level selection as an outcome-level description of origin-correlated persistence during coalescence, while separating this operational definition from causal ecological or biochemical mechanisms (Reviewer 2).
- We refined the connection between model and experiment by distinguishing the phenomenological gLV interaction-strength parameter from empirical readouts of invasion resistance, net interaction intensity, and environmental feedbacks across nutrient conditions (Reviewers 2 and 3).
- We revised key figure panels: Fig. 1B now states the cosine-similarity calculation and Euclidean-norm normalization explicitly; Fig. 2C now more clearly shows the assembly-driven decrease in within-community interaction strength; and Supplementary Fig. 3 now shows post-assembly interaction matrices ordered by parental-community origin (Reviewers 1 and 3).
- We added metric and null-model analyses, including continuous similarity measures, classification-boundary sensitivity, alternative similarity metrics, and coalescence-pair-specific additive-null comparisons (Reviewers 1–3).
- We added controls and alternative-mechanism analyses, including biomass imbalance, pool-size/richness effects, dominant-species artifacts, pH-mediated feedback, and alternative interaction structures (Reviewers 1–3).
- We narrowed the natural sample-derived community claims by emphasizing laboratory stabilization, adding taxonomic-distinctness analyses, and discussing pre-selection during stabilization as an explicit caveat (Reviewers 2 and 3).

Response to Reviewer 1

We thank the reviewer for their constructive feedback. Below we address each point in detail.

R1-1. Absolute biomass as an alternative explanation for Dominance

I found the Authors' explanation of coalescence generally convincing. However, I wonder if a simple, alternative explanation for many of the results could be that parent communities attain very different absolute densities. For example, if community A reaches 10x the overall density of community B, then even relatively rare species in community A might be initially more abundant

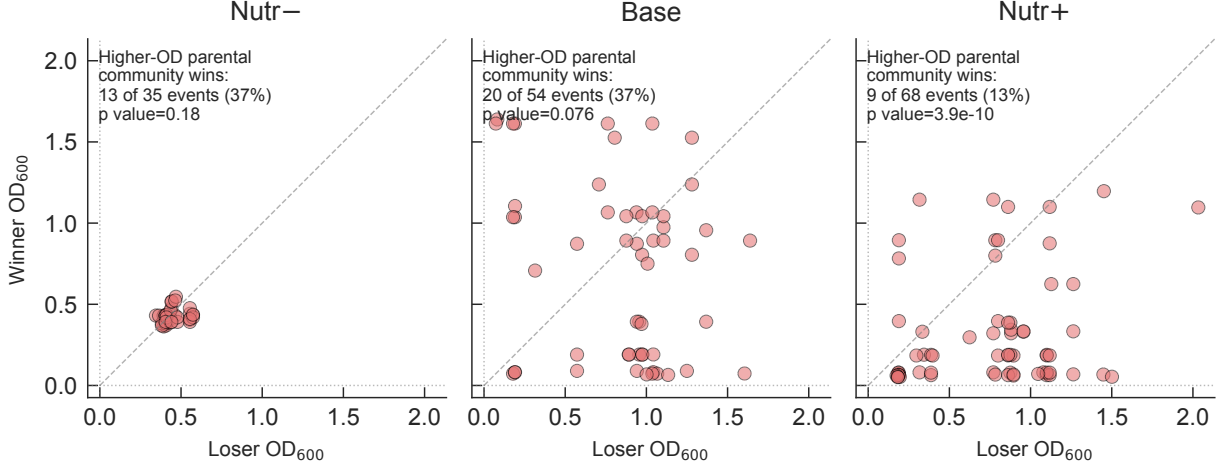
than the dominant species in community B (because communities are combined in equal volumes, line 119). This might lead to the final community being dominated by species from A simply because species survive proportionally to their initial *absolute* abundance. If I understand correctly, this scenario is different from the one tested in ED Fig. 5. High heterogeneity in absolute densities could potentially explain both the prevalence of Dominance outcomes and the difference in pairwise selection correlation between same- and cross-parent pairs. In principle, heterogeneity in absolute densities could also increase with nutrient concentration. To be clear, I have no reason to expect this pattern, but I wonder if the Authors could do more to rule it out. Since ODs were measured throughout (line 379), could those data be used to check this possibility?

We thank the reviewer for raising this important alternative explanation. We tested the suggested hypothesis by asking whether parental-community OD_{600} correlates with the parental dominance direction or with pairwise selection correlation. If Dominance were primarily driven by unequal absolute biomass at mixing, the higher-OD parental community should win more frequently, and signed ΔOD_{A-B} , the difference between the OD of parental community A and the OD of parental community B, should be positively associated with PDI. We observed the opposite pattern. In Base, the higher-OD parental community wins only 37% of Dominance events, and the continuous relationship between signed ΔOD_{A-B} and PDI is weak. This opposite-to-prediction pattern becomes strongest in Nutr+, where Dominance is most frequent: the higher-OD parental community wins only $9/68 = 13.2\%$ of Dominance events ($p = 3.9 \times 10^{-10}$), and signed ΔOD_{A-B} is strongly negatively correlated with PDI ($\rho = -0.60$, $p = 4.0 \times 10^{-10}$; Response Fig. R1-1A,B). Nutr- is less informative for this directional test because both OD differences and Dominance frequency are smaller. Thus, this directional analysis argues against parental-community biomass differences as an explanation for Dominance direction.

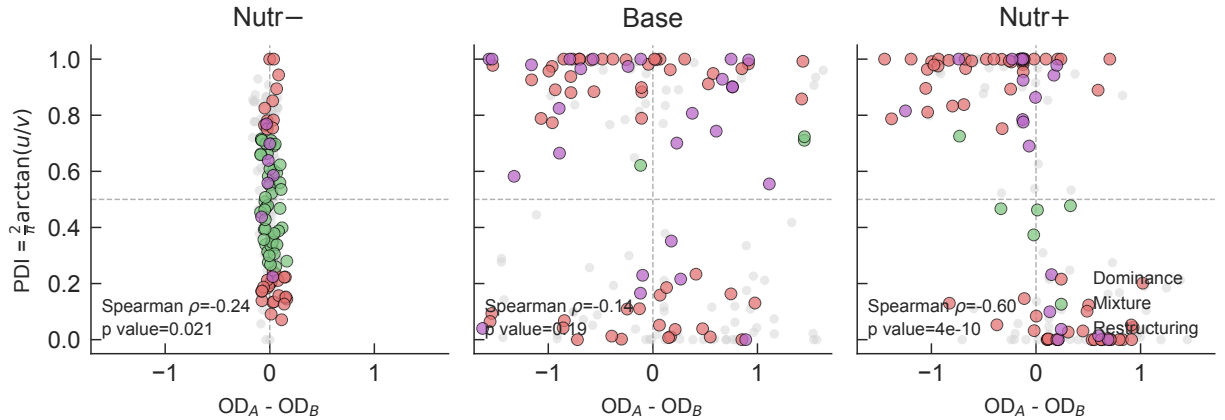
This density-based explanation also predicts that higher-OD parental communities should show stronger within-community pairwise selection correlations. To test this, we binned parental communities into low, mid, and high OD tertiles within each medium and computed the pairwise selection correlation among species from the same parental community in each OD bin (Response Fig. R1-1C). The cross-community correlation is not re-binned by OD and is shown as a medium-level reference. Within-community pairwise selection correlation does not show a consistent increase with parental-community OD; instead, the pattern is not consistent across media. In Nutr+, the slight negative OD trend mirrors the negative signed ΔOD_{A-B} -PDI association in the directional analysis above, rather than the positive trend expected if higher biomass drove winning. Overall, this analysis implies that differences in parental-community biomass do not explain Dominance direction or the nutrient-dependent pattern of correlated species fates.

We added a concise note to the Results and present the corresponding biomass and richness controls in Supplementary Note 5 and Supplementary Figs. 13–15, 45. The Results now state:

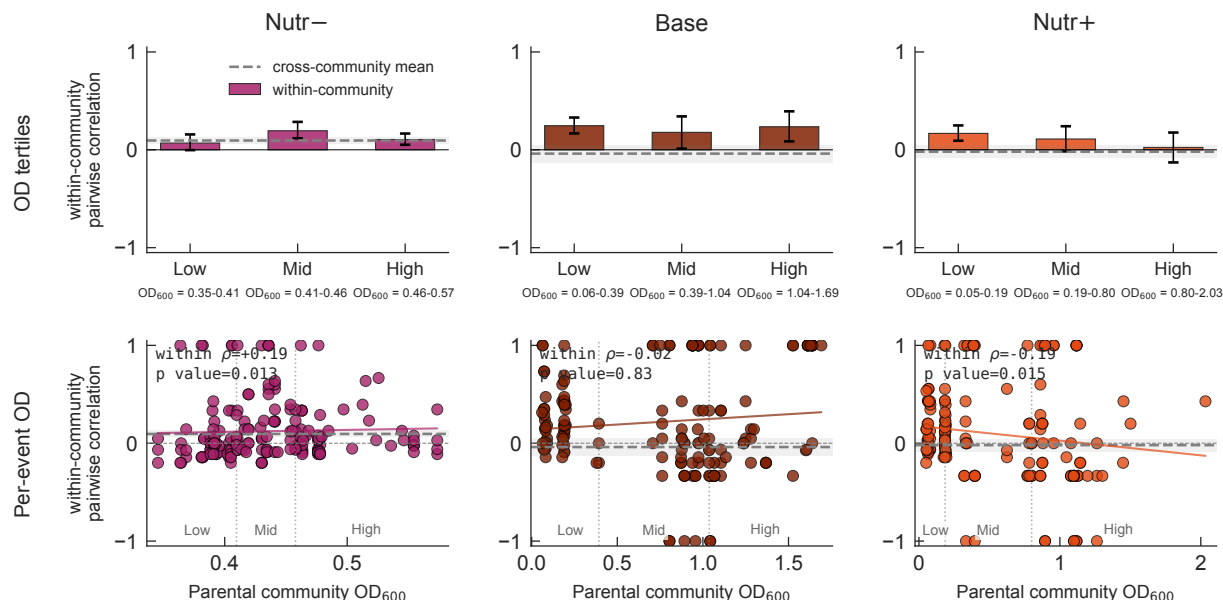
“We next tested whether nutrient-dependent changes in parental-community properties, including parental-community biomass heterogeneity and initial richness effects, could explain this trend; however, Dominance outcomes were not biased toward the higher-biomass parental community, and these factors did not account for the nutrient-dependent increase in Dominance or correlated species fates (Supplementary Note 5 and Supplementary Figs. 13–15, 45).”



Response Fig. R1-1A. Higher parental-community OD does not predict the winner in Dominance events. Winner OD vs loser OD scatter is shown per medium; dashed line is $y = x$. Points above the line are cases where the winning parental community had higher OD than the losing parental community. The higher-OD parental community wins only 37%, 37%, and 13% of Dominance events in Nutr-, Base, and Nutr+ respectively; the strongest opposite-to-prediction pattern occurs in Nutr+. The displayed p value is from a two-sided binomial test against the null expectation that the higher-OD parental community wins 50% of Dominance events.



Response Fig. R1-1B. Continuous OD differences do not support the prediction that the higher-OD parental community wins. Signed ΔOD_{A-B} ($OD_A - OD_B$) vs PDI is shown per medium. PDI is the Parental Dominance Index defined in the manuscript, $PDI = (2/\pi) \arctan(u/v)$, with PDI defined as 1 for the boundary case $v = 0$ and $u > 0$. Spearman ρ is the rank correlation between signed ΔOD_{A-B} and PDI; the displayed p value is from a two-sided Spearman rank-correlation test of the null hypothesis of no monotonic association. The relationship is weak in Base and strongly negative in Nutr+ ($\rho = -0.60$, $p = 4 \times 10^{-10}$), opposite the sign predicted if higher OD caused a parental community to win. Gray points are reflections included for symmetry.



Response Fig. R1-1C. Higher parental-community OD does not produce a consistent increase in within-community pairwise selection correlation. Columns show Nutr-, Base, and Nutr+, respectively. **Top row:** parental communities are grouped into low, mid, and high within-medium OD tertiles; smaller text below the group labels shows the OD_{600} range of each tertile. Bars show within-community pairwise selection correlation with bootstrap 95% CIs; the dashed gray line and shaded band show the medium-level mean cross-community correlation and its bootstrap 95% CI. **Bottom row:** per-event parental-community observations plotted by OD and within-community pairwise selection correlation; vertical dotted lines mark the low/mid/high OD boundaries used in the top row. Spearman ρ and p values are shown in each panel; p -values are from two-sided Spearman rank-correlation tests of the null hypothesis of no monotonic association between OD and within-community pairwise selection correlation.

R1-2. Does pH mismatch predict Dominance?

To support the hypothesis that the top-down regime emerges because dominant species strongly modify the pH of the ecosystem, the Authors show that the predictive power of the parent community pH increases in the Nutr+ condition when looking at coalescence between communities where one parent is acidic and one is alkaline (lines 261, ED Fig. 8). It seems like an even more straightforward prediction would be that Dominance should become more likely when combining parent communities with qualitatively different pH (e.g. alkaline and acidic) compared to cases where both parents are either acidic or alkaline. Is this the case?

We thank the reviewer for suggesting this direct test. We separated two related questions: whether pH contrast increases the probability of a Dominance-class outcome, and whether pH contrast predicts which parent wins within acidic-alkaline pairs.

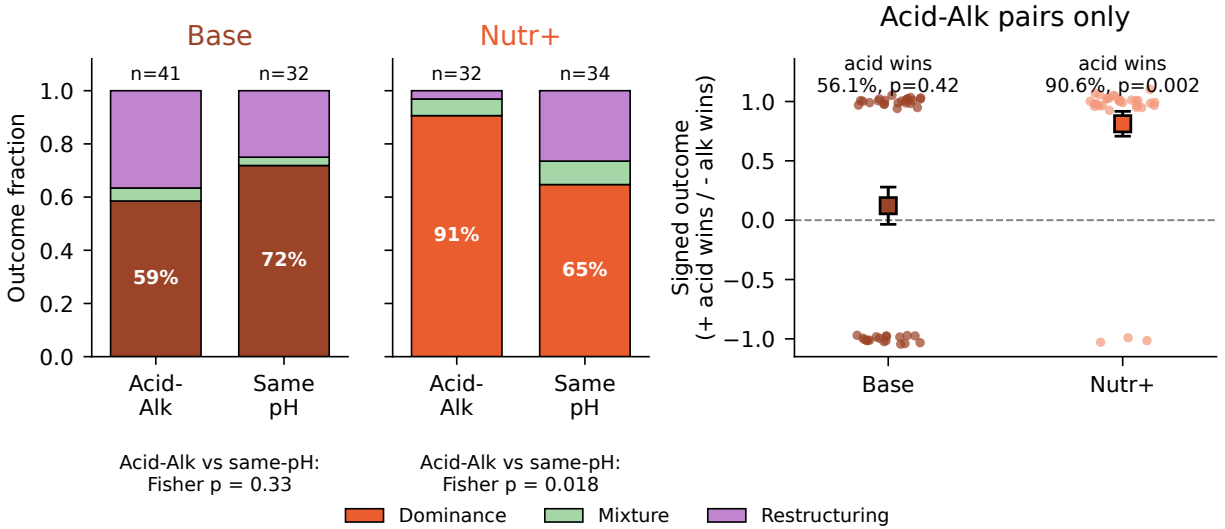
Acidic-alkaline pairings were more likely to produce a Dominance-class outcome than same-pH pairings in Nutr+: 29/32 acidic-alkaline events (90.6%) were classified as Dominance, compared with 22/34 same-pH events (64.7%, OR = 5.27, Fisher p = 0.018). In Base medium, acidic-alkaline pairings were not enriched for Dominance (24/41, 58.5%) relative to same-pH pairings (23/32, 71.9%,

OR = 0.55, Fisher $p = 0.33$). Thus, pH contrast is associated with higher Dominance frequency in Nutr+ under this comparison, but not in Base, and same-pH Nutr+ pairings still frequently produce Dominance. This supports pH contrast as a contributor to the nutrient-rich top-down regime rather than as a complete explanation for Dominance across all coalescence events (Response Fig. R1-2 and Supplementary Fig. 46).

This complements the winner-direction analysis already reported in the manuscript: acidic parental communities dominated 29/32 Nutr+ acidic-alkaline events (90.6%), compared with 23/41 Base events (56.1%, Extended Data Fig. 7).

We revised the overall claim in Results §2.5 to make clear how the pH-contrast analysis bears on this question. The revised Results paragraph states, in part:

“In coalescence events between acidic ($\text{pH} < 6.5$) and alkaline ($\text{pH} > 7.5$) parental communities, the acidic community dominated in only 56% of cases in Base medium ($n = 41$) and in 91% of cases in Nutr+ medium ($n = 32$) (Extended Data Fig. 7). The difference between Base and Nutr+ proportions was significant (two-sided Fisher exact test, $p = 0.0015$), consistent with the hypothesis that high nutrients amplify metabolic activity, thereby intensifying pH modification and excluding pH-sensitive species (Ratzke and Gore, 2018; Ratzke et al., 2020). Together, the dominant-species and strict pH-contrast analyses suggest that Nutr+ Dominance may reflect a top-down regime, in which dominant taxa and associated environmental feedbacks help determine winner identity. However, this pH-based analysis addresses winner direction in high-contrast acidic-alkaline pairings and does not by itself account for the full Dominance pattern across all coalescence events. Dominance in Base is less directly explained by the measured dominant-species and pH effects tested here.”



Response Fig. R1-2. Strict parental-community pH contrast is most informative in Nutr+. Base and Nutr+ panels show stacked Dominance, Mixture, and Restructuring fractions for strict acidic-alkaline and strict same-pH pairings. Parental communities were classified as acidic when pH < 6.5 and alkaline when pH > 7.5. Intermediate-pH pairings were excluded. Statistical tests compare Dominance with pooled Mixture and Restructuring outcomes. Strict acidic-alkaline pairings were enriched for Dominance in Nutr+ (29/32 versus 22/34 events, OR = 5.27, Fisher $p = 0.018$), but not in Base medium (24/41 versus 23/32 events, OR = 0.55, Fisher $p = 0.33$). The Acid-Alk pairs-only panel shows winner direction within strict acidic-alkaline pairings: the acidic parental community dominated 29/32 Nutr+ events (90.6%), compared with 23/41 Base events (56.1%, Extended Data Fig. 7). Dots show individual binary winner-direction outcomes, and squares show mean $\pm$ s.e.m.

R1-3. Circularity in Fig. 5C, PDI excluding dominant species

In Fig. 5C, it seems like there is some risk of circularity if the dominant species are included in the calculation of the PDI. Does the positive correlation under Nutr+ conditions hold up if the dominant species are excluded from this calculation?

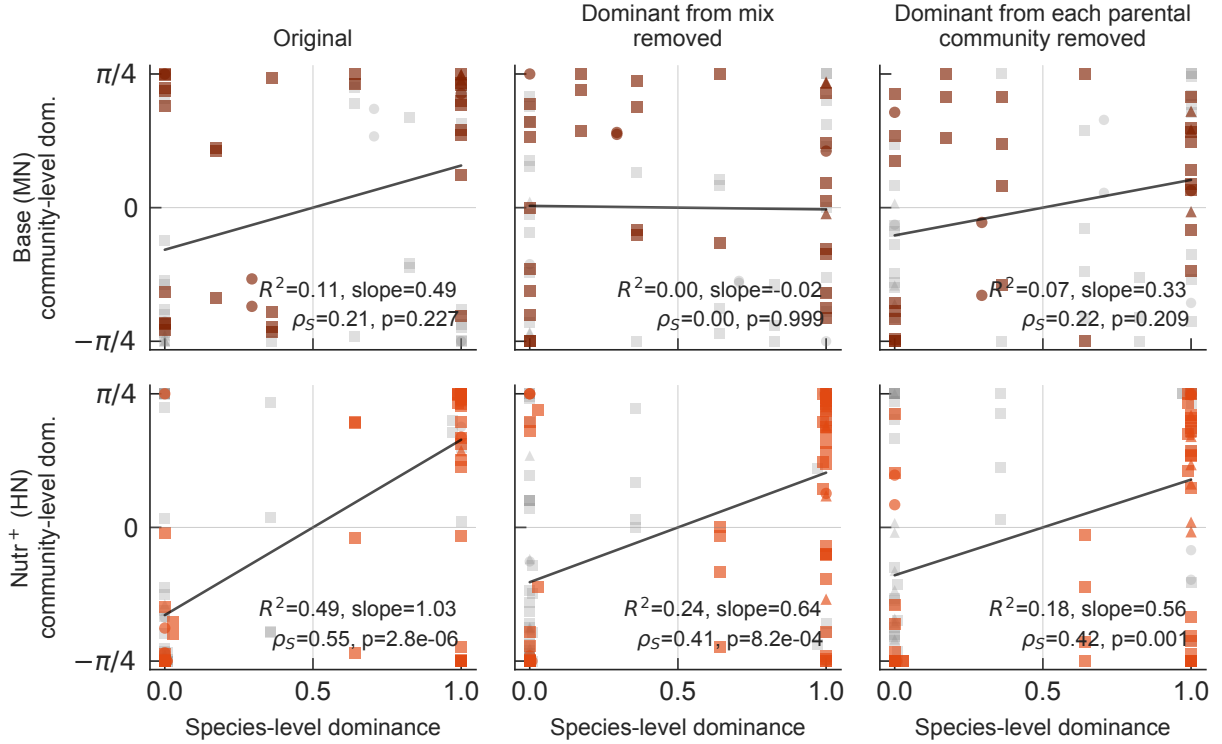
We agree that this is an important circularity check. We used two removal strategies to avoid biasing the result toward the dominant species in the coalesced community. In the first, we removed the most abundant species in the coalesced community from all three community composition vectors; in the second, stricter test, we identified the most abundant species in each parental community and removed those taxa from all three composition vectors before recalculating PDI. We followed the same PDI calculation and filtering criteria as in the original Fig. 5C analysis: events are included if they are classified as Mixture or Dominance before dominant species removal.

This analysis reveals that our core conclusion remains supported: dominant-species pairwise interactions remain associated with the recalculated community-level PDI in Nutr+, but not in Base. In Base, R^2 drops from 0.11 to 0.00/0.07 and the slope from 0.49 to $-0.02/0.33$ (Original / coalesced-community dominant removed / parental-community dominants removed). In Nutr+, R^2 drops from 0.49 to 0.24/0.18 and the slope from 1.03 to 0.64/0.56 (Response Fig. R1-3). Using indepen-

dent, non-reflected events as the primary inference, the Spearman correlation between dominant-species pairwise score and community-level score remains significant after both removal strategies: $\rho = 0.41$ ($p = 8.2 \times 10^{-4}$) after removing the coalesced-community dominant species and $\rho = 0.42$ ($p = 1.3 \times 10^{-3}$) after removing the dominant species from both parental communities. Thus, dominant taxa contribute substantially to the Nutr+ signal, but the Nutr+ association does not disappear when those taxa are excluded. In contrast, Base does not retain a significant Spearman correlation after removal ($p = 1.00$ and 0.21).

We added a concise note to Results §2.5 and present the primary analysis in Supplementary Fig. 16. The Results now state:

“The Nutr+ predictability trend remained significant after recalculating PDI with dominant species excluded (Spearman rank-correlation tests: $p = 8.2 \times 10^{-4}$ after removing the coalesced-community dominant species and $p = 1.3 \times 10^{-3}$ after removing the two parental-community dominant species; Supplementary Fig. 16), indicating that it is not solely a dominant-species artifact in the PDI calculation.”



Response Fig. R1-3. Nutr+ predictability persists after dominant-species removal, arguing against a dominant-species artifact in the PDI calculation. Per-medium scatter plots are shown with no pooling between Base and Nutr+. Rows: Base (MN, brown) and Nutr+ (HN, orange). Columns: original Fig. 5C calculation; dominant species from the coalesced community removed from all three composition vectors; and dominant species from each parental community removed from all three composition vectors before recalculating PDI. Colored points are independent coalescence events; gray points are reflected counterparts plotted for symmetry. The third column is the stricter test. R^2 and slope annotations are computed on the reflected plotting data, as in Fig. 5C, and are used as descriptive summaries. Spearman ρ and p are computed on the independent, non-reflected events; Spearman p -values are from two-sided rank-correlation tests of the null hypothesis of no monotonic association.

R1-4. Pool-size effects

I wondered if the Authors could provide slightly more detail on the effect of the initial pool size. Is it similarly true that there is no significant effect of initial richness in the experimental communities? By eye, it looks like Dominance might be more likely for the 24 species pool, but it is hard to tell by squinting at Fig. 1E. It might also be interesting to see the survival ratio as a function of initial richness for the experimental communities.

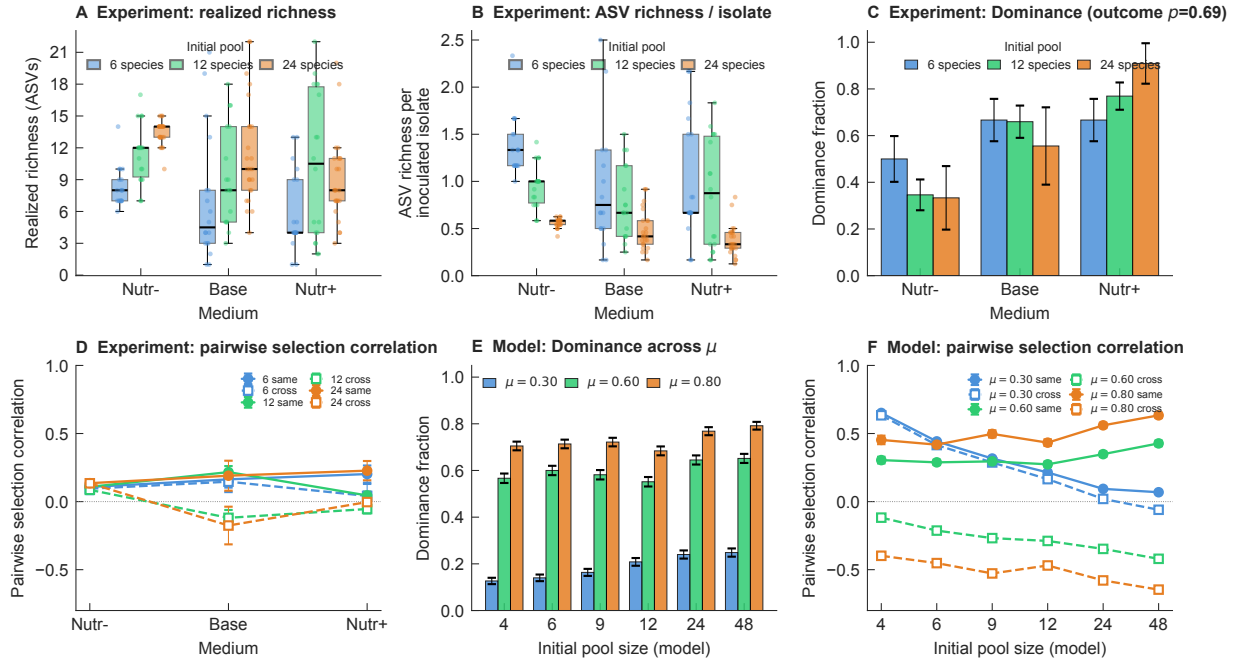
We did not detect a significant effect of pool size in the experiment, and we thank the reviewer for prompting this more direct check. Response Fig. R1-4C shows Dominance frequency by initial pool size within each nutrient condition. Across nutrient conditions, the full three-category outcome distribution did not vary significantly with pool size ($\chi^2 = 2.24$, $p = 0.69$), and binary Dominance frequency was also not significantly associated with pool size within individual nutrient conditions

(Dominance vs non-Dominance tests: Nutr- $p = 0.39$, Base $p = 0.82$, Nutr+ $p = 0.27$). Thus, although initial pool size affects realized richness and ASV richness per inoculated isolate (panels A,B), these assembly-richness effects do not translate into a detectable pool-size dependence of Dominance frequency with the available sampling.

I also wondered if the Authors could look more closely at why there is no effect in the model. Perhaps by plotting the realized richness and interaction strength in parent communities as a function of initial richness?

As a complementary model check, we performed a gLV pool-size ablation study at low, intermediate, and high values of the interaction-strength parameter μ , varying the initial pool size from 4 to 48 species per parental community (100 replicate assemblies per μ and pool size). Consistent with the reviewer’s intuition, reconstructing the simulated interaction matrices showed that assembly keeps within-community interaction coefficients lower than between-community coefficients across pool sizes. The outcome-level model check gives the same interpretation: at fixed μ , Dominance frequency changes only modestly with pool size compared with the strong effect of μ itself (panel E), whereas the same-parental-community versus cross-parental-community pairwise selection correlation gap increases with pool size (panel F). Thus, pool size can modulate the strength of correlated species fates, but it does not fully explain the primary interaction-strength-dependent transition in Dominance frequency.

We added this analysis to Supplementary Note 5, “Alternative explanations for the nutrient-dependent Dominance,” and to Supplementary Fig. 13. The new supplementary text states that pool-size variation affects assembly richness but does not explain the Dominance-frequency pattern, and that the model pool-size ablation supports the same conclusion.



Response Fig. R1-4. Dominance frequency is primarily associated with nutrient condition or model interaction strength, rather than initial pool size. **A:** Realized parental richness. **B:** ASV richness per inoculated isolate, used as an assembly-retention proxy and computed as realized parental ASV richness divided by inoculated species-pool size; this ratio can exceed one because the numerator is thresholded ASV richness rather than a count restricted to the inoculated isolates. **C:** Experimental Dominance fraction by medium and initial pool size; the displayed χ^2 test refers to the full three-category outcome distribution across pool sizes. **D:** Experimental pairwise selection correlation metric grouped by initial pool size. **E:** Model Dominance fraction grouped by μ . **F:** Model pairwise selection correlation metric across pool sizes 4–48.

R1-5. Similarity-metric robustness claim

The Authors write (line 134) that “This pattern of Dominance as the most frequent outcome was robust across variants of similarity metrics”. I liked the Author’s approach to measuring outcomes, and I don’t advocate for changing it, but this claim seems a little misleading. Two of four alternative similarity metrics gave different results, with Dominance being the *least* likely outcome under Jensen–Shannon and Jaccard.

We agree that our original robustness claim was too broad and could be misleading: the Dominance pattern is recovered by Euclidean distance and Bray–Curtis dissimilarity, but not under Jaccard index or Jensen–Shannon divergence (Extended Data Fig. 2). This pattern is consistent with the fact that the metrics emphasize different aspects of community composition: abundance-weighted parental-similarity metrics such as vector decomposition, Euclidean distance, and Bray–Curtis dissimilarity are more sensitive to quantitative dominance in relative abundance, whereas Jaccard reflects presence/absence retention and Jensen–Shannon evaluates divergence across the full abundance distribution. Thus, a coalesced community can be skewed toward one parental community in abundance space while still retaining taxa from both parental communities, or while redistributing

subdominant taxa in ways that make it less skewed toward one parental community under Jaccard or Jensen–Shannon.

We revised Results §2.1 to state the mixed metric result directly:

“Alternative metric analysis revealed that Dominance remained the most frequent outcome under abundance-weighted compositional metrics, including Euclidean distance and Bray–Curtis dissimilarity, but not under Jaccard index or Jensen–Shannon divergence, which emphasize species-identity retention and full-distribution divergence, respectively (Extended Data Fig. 2).”

We also revised the Extended Data Fig. 2 caption to explicitly discuss the divergent metrics:

“Dominance is recovered by abundance-weighted parental-similarity metrics, including the primary similarity-map classification, Euclidean distance, and Bray–Curtis dissimilarity, but not by Jaccard index or Jensen–Shannon divergence. This divergence reflects the different biological questions asked by each metric: abundance-weighted metrics quantify whether the coalesced community quantitatively resembles one parent more than the other, whereas Jaccard index measures presence/absence retention and Jensen–Shannon divergence measures differences across the full abundance distribution. Thus, Jaccard and Jensen–Shannon are best interpreted as complementary metric choices that emphasize different features of community composition.”

R1-6. Gray reflected points in figures

It took me quite a while to realize that the gray points in Figs. 1E, 4C, 5C, and 6B are simply a reflection of the colored points. Please indicate this clearly, or perhaps just remove these.

We retained the reflected points and updated the captions of Figs. 1E, 4C, 5C, and 6B to state explicitly that the gray points are reflections:

“Gray points represent the symmetric counterpart of each coalescence event (community B versus A).”

R1-7. Interaction matrix after assembly

In Fig. 2A, it might be helpful to show the matrix of interaction coefficients after assembly of the parent communities (with visible block structure).

We agree that this visualization is helpful. We added the requested post-assembly interaction-matrix analysis as Supplementary Fig. 3, where post-assembly survivor matrices are ordered by parental-community origin, and kept Fig. 2A focused on the simulation workflow. We also revised Fig. 2C and the Results text to emphasize the assembly-driven decrease in within-community interaction strength.

R1-8. Fig. 2D visualization

Fig. 2D, I was somewhat confused by the relationship between points and squares (means?). Visually, the squares do not look like they are near the means of the points. Please clarify this for readers. Additionally, the gray horizontal bars are not explained in the main text unless I missed it.

In the simulation panel, the dots show a stratified subset of per-event pairwise selection correlations for visual legibility, whereas the squares and error bars show the mean $\pm$ s.e.m. across all simulated coalescence events. We revised the Fig. 2D caption to specify (1) what the dots represent, (2) what the squares and error bars represent, and (3) that the gray horizontal lines are the random-selection baseline from a parental-affiliation-shuffle null model:

“Individual dots show per-event pairwise selection correlations; in the simulation panel, 100 stratified events per group are displayed for legibility. Squares with error bars show mean $\pm$ s.e.m. across all coalescence events ($n = 1,200$ per group in simulations and all valid experimental events in the experimental panel). Gray horizontal lines indicate the random selection baseline from a null model in which parental-affiliation labels are shuffled (see Supplementary Information).”

R1-9. Emphasize “cohesion without cooperation”

Discussion, lines 342–345: I think it is worth emphasizing how interesting and perhaps surprising it is to find “community-level cohesion without cooperation” as the paper by M. Tikhonov has it. This finding will be counterintuitive to many readers, even though the mechanism is made clear.

We agree with the reviewer that this is a central and counterintuitive implication of the study. We added a dedicated short paragraph to the Discussion that frames the result in these terms and situates this point in relation to Tikhonov’s prediction and random Lotka–Volterra theory.

“Notably, and perhaps counterintuitively, our model shows that community-level selection-like patterns can emerge without explicitly cooperative interactions between species, consistent with predictions from Tikhonov’s resource-competition models (Tikhonov, 2016) and random Lotka–Volterra theory (May, 1972; Grilli et al., 2017). Community-level selection is often attributed to cooperative mechanisms such as cross-feeding or division of labor. By contrast, the model suggests that assembly history and competitive interactions can be sufficient to generate correlated species fates during coalescence. Thus, cooperative and competitive mechanisms may represent distinct routes to community-level selection, and understanding how these routes interact remains an important direction for future work.”

R1-10. Means missing in Extended Data Fig. 5C

In ED Fig. 5C, are means missing?

We thank the reviewer for pointing this out. The means are shown in Extended Data Fig. 5C as square markers with error bars, but they were not sufficiently clear in the previous version. We revised Extended Data Fig. 5 so that the mean markers and error bars are more visible, and the caption now states that individual dots show per-event correlations whereas squares with error bars show mean $\pm$ s.e.m.

R1-11. Incorrect Extended Data reference in the SI

On p. 9 in the SI, I believe the sentence “Simulations were performed with parental community sizes ranging from 4 to 48 species per community (Extended Data Fig. 5)” should refer to ED Fig. 4.

We thank the reviewer for catching this error and corrected the SI reference.

Response to Reviewer 2

We thank the reviewer for their thoughtful and constructive comments, which have significantly improved the conceptual clarity of the manuscript. Below we address each point.

R2-1. Nutrient enrichment versus interaction strength

The central claim of the paper is that increasing nutrient supply strengthens interactions, which in turn drives a shift toward community-level selection. While this is an appealing and intuitive interpretation, it may be overly simplified.

Nutrient enrichment simultaneously affects multiple ecological processes, including:

- species' carrying capacities
- metabolic rates
- environmental modification (e.g. pH shifts)
- the balance of competition and facilitation

Indeed, the manuscript itself shows that pH modification by dominant taxa may contribute to winner identity in the nutrient-rich regime, while not serving as a sole explanation for Dominance frequency. This suggests that the observed transition may reflect changes in environmental feedbacks and dominance structure, rather than a simple increase in pairwise interaction strength.

This raises an important conceptual point:

Is the observed shift in coalescence outcomes driven by increased interaction strength per se, or by other effects of resource enrichment?

We therefore suggest reframing this result more broadly in terms of changes in *interaction intensity and environmental feedbacks*, rather than interpreting nutrient enrichment as a direct proxy for stronger interactions. This would align well with recent work highlighting the role of environmentally mediated interactions in microbial communities (Goldford et al., 2018; Estrela et al., 2021).

We agree that this is an important conceptual question to clarify. In our work, the gLV interaction-strength parameter is framed as a phenomenological parameter for ecological coupling within the model, while the nutrient manipulation in Fig. 4 changes nutrient availability and is interpreted through its effects on invasion resistance, net interaction effects, and environmental feedbacks. We therefore use failed pairwise invasion frequency as an operational measure of invasion resistance, in order to approximate how net interaction effects change across nutrient conditions, rather than as a direct measurement of one biochemical mechanism or a literal estimate of every pairwise Lotka–Volterra coefficient. Multiple mechanisms, including resource competition, increased metabolic activity, pH-mediated environmental feedbacks, and changes in the competition–facilitation balance, can contribute to the observed nutrient-dependent change in invasion resistance. This framing is consistent with prior work on nutrient-mediated microbial interactions, environmentally mediated community assembly, and nutrient-concentration-dependent changes in microbial community diversity and structure (Ratzke and Gore, 2018; Ratzke et al., 2020; Goldford et al., 2018; Estrela et al., 2021; Dal Bello et al., 2026; Duan et al., 2025).

From a consumer–resource perspective, it is also not mathematically self-evident that increasing resource supply monotonically increases effective pairwise interaction strengths. In our recent theoretical work (Duan et al., 2025), we show that when mapping mechanistic consumer–resource dynamics to effective Lotka–Volterra coefficients, the external supply rate can cancel out under broad conditions, provided that resources are not strictly limiting and metabolic strategies remain fixed.

Under such conditions, resource enrichment increases total biomass but does not necessarily increase mean per-capita competition strength. Instead, enrichment restructures the interaction network through changes in equilibrium abundances and environmental feedbacks. This distinction is important, as it suggests that the observed transition may reflect a reorganisation of interaction structure rather than a simple scaling of interaction strength.

A brief discussion of these issues would significantly strengthen the conceptual clarity of the manuscript.

The apparent discrepancy comes from how the interaction coefficient is defined. The consumer–resource result discussed by Duan, Pawar and colleagues concerns coefficients derived mechanistically from resource dynamics as effective *per-capita* terms in a population-growth equation. In contrast, our gLV coefficients are coarse-grained effective inhibitory coefficients in a per-capita growth equation within a phenomenological abundance model, and our nutrient-gradient experiment uses pairwise invasion assays to quantify empirical invasion resistance. We therefore do not interpret nutrient enrichment as a direct or monotonic increase in those mechanistic *per-capita* Lotka–Volterra coefficients. As the reviewer correctly notes, resource enrichment can increase total biomass without simply scaling the underlying *per-capita* coefficients. Thus, the nutrient-gradient result should be interpreted as an increase in empirical invasion resistance, not as a claim that nutrient supply directly or monotonically scales every effective *per-capita* interaction coefficient.

Results §2.4 (“Nutrient-dependent invasion resistance and environmental feedbacks shift coalescence outcomes”) now states, in part:

“Following prior work showing that nutrient concentration can alter microbial competition and that pH-mediated environmental feedbacks can drive bacterial interactions (Hu et al., 2022; Ratzke and Gore, 2018; Ratzke et al., 2020; Hu et al., 2025), we conducted additional coalescence experiments by removing glucose and urea from the Base medium or increasing their concentrations (Methods). This yielded two additional media conditions (Fig. 4A): Nutr– (no added glucose/urea) and Nutr+ (high supplementation). To approximate how net interaction effects change across nutrient conditions, we measured invasion resistance using pairwise invasion assays among the 12 most abundant isolates (95:5 initial frequency; Methods, Supplementary Figs. 7–9).”

R2-2. Alternative explanations for community-level selection

The manuscript interprets dominance of one parental community and correlated persistence of its taxa as evidence for community-level selection. We find this interpretation interesting and potentially insightful, but it is not uniquely diagnostic on its own.

Similar patterns could arise from:

- shared environmental filtering
- correlated traits within parental communities

- environmental modification (e.g. pH tolerance)

In particular, the nutrient-rich regime appears consistent with strong environmental filtering imposed by a small number of dominant taxa, which may not require invoking selection acting at the level of the community as a unit.

We therefore recommend clarifying the interpretation of “community-level selection,” and explicitly discussing alternative mechanisms that could generate similar patterns. This distinction has been discussed in previous work on community coalescence and microbial assembly (Mansour et al., 2018; Rillig et al., 2015).

We thank the reviewer for raising this important point. We agree that Dominance and correlated persistence should not be interpreted as uniquely identifying a single underlying mechanism. We have therefore revised the manuscript to present Dominance as outcome-level evidence for community-level selection, while defining community-level selection operationally rather than as one exclusive causal process. In the Discussion, we now state that community-level selection describes origin-correlated persistence during coalescence and that this operational definition does not by itself identify causal ecological or biochemical mechanisms, which may vary across systems.

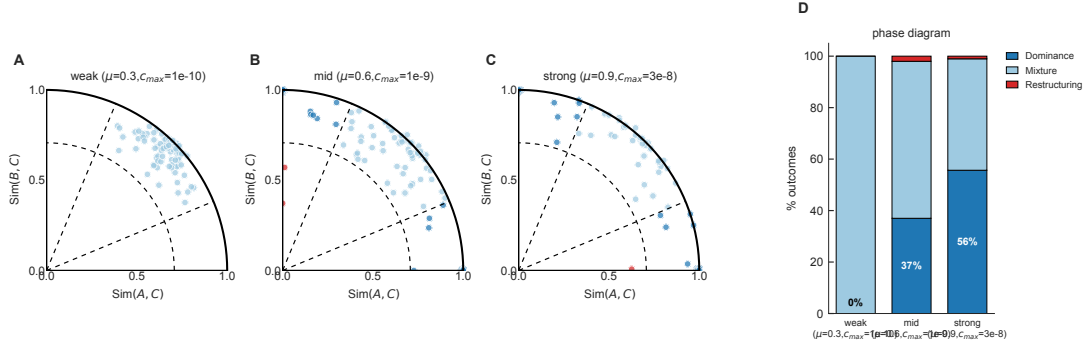
We made this distinction explicit in Results §2.1 (“Community-level selection is prevalent in microbial coalescence”):

“Crucially, Dominance provides outcome-level evidence for community-level selection: one parental community largely displaces the other while preserving its compositional structure.”

At the start of the Discussion paragraph on broader ecological settings, we now make this distinction explicit:

“Accordingly, we use community-level selection as an outcome-level description of origin-correlated persistence during coalescence. This operational definition does not by itself identify the causal ecological or biochemical mechanisms, which may vary across systems.”

As an additional plausibility check for environmental feedback, we ran a Ratzke–Gore-style pH-feedback model (Ratzke and Gore, 2018), which produces asymmetric replacement over an intermediate range of feedback strength (Response Fig. R2-2 and Supplementary Fig. 43). We agree that pH-mediated environmental modification is a plausible mechanism for coalescence outcomes, and we do not claim that the phenomenological gLV model is the unique explanatory framework. One observation from this comparison is that the pH-only model rarely produces Restructuring outcomes ($\sim 1\%$ across feedback strengths; 3/297 simulated events), whereas Restructuring is observed at substantially higher rates both in our experiments ($\sim 8\text{--}31\%$ across nutrient conditions) and in the phenomenological gLV model ($\sim 22\%$ across μ values). This comparison does not rule out pH feedback; instead, it shows that pH feedback alone is unlikely to explain the full outcome distribution. As discussed in R2-1, pH-mediated environmental effects can be included implicitly in the coarse-grained gLV coefficients as part of the net inhibitory effect of one species on another, although pH itself is not an explicit state variable in the model.



Response Fig. R2-2. Outcome distribution of a Ratzke–Gore-style pH-feedback model across feedback strengths. Per-event outcomes in the similarity plane (left, three feedback strengths) and stacked outcome fractions (right) for the pH-feedback model. The model produces asymmetric replacement (Dominance) over an intermediate feedback range, supporting pH-mediated environmental modification as a plausible mechanism contributing to coalescence outcomes. Restructuring outcomes are rare across all feedback strengths ($\sim 1\%$; 3/297 simulated events), in contrast to the substantial Restructuring fractions observed experimentally ($\sim 8\text{--}31\%$) and in the phenomenological gLV model ($\sim 22\%$).

In addition, several related comments raise alternative theoretical models and complementary controls, which we address in the corresponding response sections: parental-community biomass effects (R1-1), pH-mediated filtering and winner direction (R1-2, plus the R2-2 and Supplementary Fig. 43 pH-feedback model above), the dominant-taxon removal control for the PDI calculation (R1-3), initial pool size and realized richness effects (R1-4 and R3-3), abundance-skew/additive-null geometric effects (R3-2 and R3-3), and facilitation or competition-only model scope (R3-4). These analyses are now organized across Supplementary Notes 1, 3, and 5: metric and null-model controls are in Supplementary Note 1, simulation-scope controls are in Supplementary Note 3, and biological alternative explanations are in Supplementary Note 5. The passive-retention alternatives are most relevant here: the denser parental community does not generally win, and we did not detect a significant overall effect of initial pool size on Dominance frequency (R1-1, R1-4).

Relatedly, we suggest clarifying the ecological meaning of “failed invasion.” At present it is primarily used as a proxy for interaction strength, but it is also naturally interpretable as invasion resistance (i.e. whether a rare invader can establish). Framing this more explicitly in terms of invasion fitness would help connect the experimental results to ecological theory and may provide additional insight into the observed patterns.

We thank the reviewer for this important clarification. We agree that failed invasion is naturally interpreted as invasion resistance and is connected to invasion fitness. We therefore revised the manuscript to frame the pairwise assay primarily as an empirical readout of invasion resistance, with invasion fitness as the relevant ecological concept. In Fig. 4B, we retained the y-axis as the measured quantity, “frequency of failed invasions,” while the panel framing identifies this as an invasion-resistance analysis. Results §2.4 now states:

“To approximate how net interaction effects change across nutrient conditions, we measured invasion resistance using pairwise invasion assays among the 12 most abundant isolates (95:5 initial frequency; Methods, Supplementary Figs. 7–9). The fraction of failed invasions increased monotonically with

nutrient supply (Nutr−: $2 \pm 1\%$; Base: $33 \pm 4\%$; Nutr+: $48 \pm 4\%$; mean $\pm$ s.e.m.; Fig. 4B), indicating that invasion resistance increases across the nutrient gradient.”

The Methods now states that

“The fraction of failed invasions served as an empirical proxy for net interaction intensity and invasion resistance (Fig. 4B).”

We distinguish this two-species assay from the pairwise selection correlation metric in R2-4.

R2-3. Continuous measures alongside categorical outcomes

The classification into Dominance, Mixture, and Restructuring is useful and provides a clear framework for summarising outcomes. However, it depends on thresholding a continuous similarity measure.

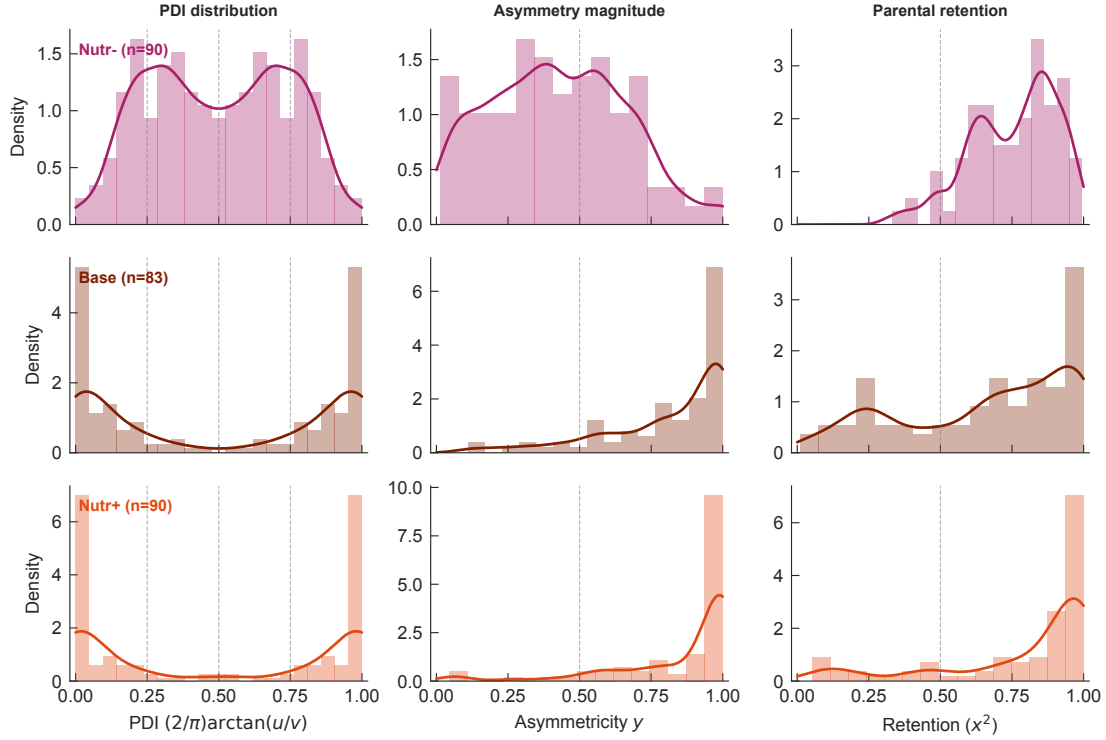
As noted in the broader literature (Mansour et al., 2018), coalescence outcomes often lie along a continuum rather than discrete categories.

We suggest that the authors:

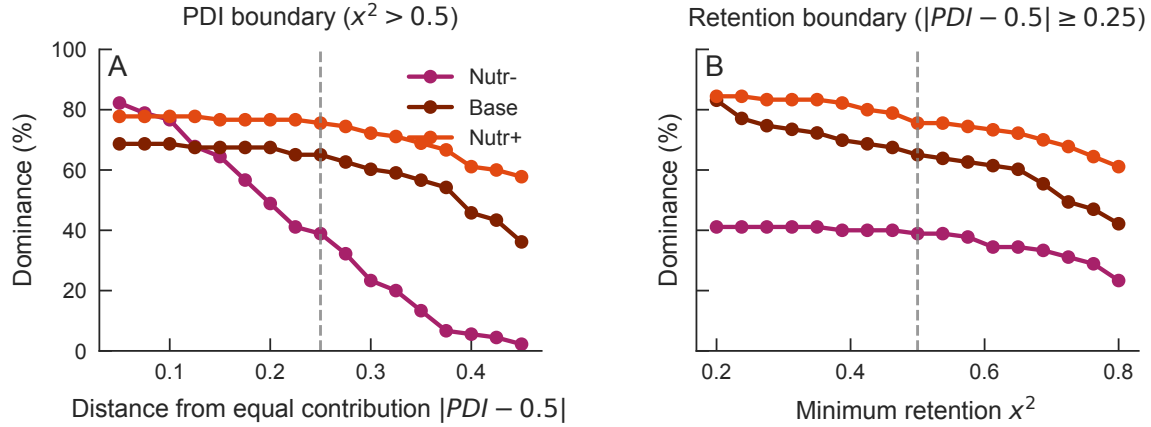
- present continuous similarity measures alongside categorical outcomes

We agree that the classification (Dominance, Mixture, Restructuring) depends on boundaries in a continuous similarity space. PDI was already introduced in Figs. 3 and 4 as a continuous measure of how strongly the coalesced community is directionally weighted toward one parental community versus the other. To make the thresholding explicit, we now present PDI together with asymmetry and retention magnitude as continuous auxiliary metrics underlying the discrete outcome classification, as shown in Response Fig. R2-3a. We added these analyses to the Supplementary Information as separate medium-specific panels: Supplementary Fig. 10 for Base, Supplementary Fig. 11 for Nutr−, and Supplementary Fig. 12 for Nutr+. These measures show where each coalescence event lies before category boundaries are applied, while the categorical labels provide the compact summary used in the main analyses.

To verify how sensitive our main nutrient-dependent result is to the classification-boundary choices, we varied the PDI and retention boundaries around the manuscript baseline (Response Fig. R2-3b). Across a broad range of boundary choices, the Dominance fraction remains ordered Nutr− < Base < Nutr+, supporting the robustness of the nutrient-dependent trend. We also added this boundary-sensitivity analysis to the Supplementary Information as Supplementary Fig. 37 and refer to it in Supplementary Note 1.



Response Fig. R2-3a. Continuous similarity measures track the same nutrient-dependent shift as the categorical classifications. Rows show Nutr-, Base, and Nutr+; columns show Parental Dominance Index (PDI), where values near 0 or 1 indicate strong asymmetry toward one parental community and values near 0.5 indicate balanced parental contributions; parental asymmetry $|2\text{PDI} - 1|$; and squared retention magnitude $r^2 = u^2 + v^2$, which measures how much of the coalesced composition is explained by the parental communities.



Response Fig. R2-3b. Boundary-sensitivity analysis supports the nutrient-dependent Dominance trend. (A) Dominance fractions after varying the PDI boundary while holding the retention boundary at $r^2 > 0.5$. (B) Dominance fractions after varying the retention boundary while holding the PDI boundary at the manuscript baseline. Dashed lines mark the manuscript baseline. This sensitivity analysis is descriptive.

In addition, the distinction between mixture and restructuring is not entirely straightforward biologically, particularly because abundance changes contribute to classification.

We suggest that the authors:

- clarify the biological interpretation of “restructuring”

We clarify that Restructuring denotes low parental retention: the coalesced community is not well explained by the parental communities or their combination. In this interpretation, Restructuring can arise when cross-community species pairings that were not present during parental assembly generate a new stable composition, especially in complex multi-species communities where abundance shifts and higher-order interactions make the final state poorly described by the parental-community basis.

Results §2.1 (“Community-level selection is prevalent in microbial coalescence”) now includes the concise Restructuring definition:

“Restructuring denotes low parental retention: the coalesced community is not well explained by the parental communities or their combination, and may reflect a new stable composition produced by previously unseen cross-community species interactions.”

We also find the comparison between dot product and Jaccard metrics in Extended Data Fig. 2 potentially very informative. The divergence between these metrics in low-nutrient environments may itself provide evidence that species-level processes dominate in that regime, and this could be discussed more explicitly.

We agree that this comparison is informative. Extended Data Fig. 2 was generated for the Base-medium outcomes, so we discuss the metric divergence in that context. Consistent with our response to Reviewer 1, we now frame the result as a difference in what each metric measures rather than as a broad robustness claim. This pattern is consistent with the fact that the metrics emphasize different

aspects of community composition: abundance-weighted parental-similarity metrics are more sensitive to quantitative dominance in relative abundance, whereas Jaccard reflects presence/absence retention and Jensen–Shannon evaluates divergence across the full abundance distribution. Thus, a coalesced community can be skewed toward one parental community in abundance space while still retaining taxa from both parental communities, or while redistributing subdominant taxa in ways that make it less skewed toward one parental community under Jaccard or Jensen–Shannon.

Results §2.1 now states the mixed metric result directly:

“Alternative metric analysis revealed that Dominance remained the most frequent outcome under abundance-weighted compositional metrics, including Euclidean distance and Bray–Curtis dissimilarity, but not under Jaccard index or Jensen–Shannon divergence, which emphasize species-identity retention and full-distribution divergence, respectively (Extended Data Fig. 2).”

Extended Data Fig. 2 caption now carries the metric-divergence interpretation:

“Dominance is recovered by abundance-weighted parental-similarity metrics, including the primary similarity-map classification, Euclidean distance, and Bray–Curtis dissimilarity, but not by Jaccard index or Jensen–Shannon divergence. This divergence reflects the different biological questions asked by each metric: abundance-weighted metrics quantify whether the coalesced community quantitatively resembles one parent more than the other, whereas Jaccard index measures presence/absence retention and Jensen–Shannon divergence measures differences across the full abundance distribution. Thus, Jaccard and Jensen–Shannon are best interpreted as complementary metric choices that emphasize different features of community composition.”

R2-4. Interpretation of pairwise selection correlation

The pairwise “selection correlation” metric is an interesting attempt to quantify lineage-level coherence, but its interpretation remains somewhat unclear.

It appears closely related to co-occurrence patterns, and it is not fully clear whether the observed correlations reflect ecological interactions, shared environmental responses, or methodological effects. We find this analysis potentially insightful, but its conceptual interpretation would benefit from clarification.

We agree that the pairwise selection correlation needed a clearer interpretation. In practice, it tests, for each coalescence event, whether species pairs from the same parental community share survival outcomes more strongly than species pairs from different parental communities. The comparison is evaluated against a within-event null in which parental-origin labels are shuffled. This controls for generic shared-fate tendencies within an event, so the measured enrichment specifically asks whether shared fate is associated with parental-community origin. We use this enrichment as evidence for community-level selection, while recognizing that the metric does not identify the causal route; direct competition, shared environmental tolerance, and environmental modification can all generate such structure.

Results §2.2 now defines the metric directly:

“We quantified this effect using pairwise selection correlation, which tests across coalescence events whether species pairs from the same parental community share survival outcomes (both persist or both go extinct) more strongly than species pairs from different parental communities (Supplementary Information; Fig. 2D).”

We suggest adding a short paragraph linking this metric more explicitly to invasion fitness in a gLV framework. In that context, invasion fitness corresponds to the growth rate of a species when rare in a resident community. The pairwise invasion assays can then be interpreted as empirical approximations of invasion fitness in two-species systems, while coalescence outcomes reflect correlations in invasion success across many species.

This would provide a unifying theoretical framework linking pairwise assays, community coalescence, and the notion of community-level selection.

We agree that this point needs clarification, especially because “pairwise” refers to two different analyses in the manuscript. As discussed in R2-2, the 95:5 pairwise invasion assay measures two-species growth outcomes and is used primarily as an empirical approximation of invasion resistance under each nutrient condition. By contrast, the pairwise selection correlation metric is computed from community coalescence outcomes and asks whether species fates are more strongly coupled within parental communities than between parental communities.

In the gLV framework, these are related but distinct readouts of invasion success and persistence. Pairwise invasion assays approximate rare-invader establishment in two-species resident backgrounds, whereas selection correlation asks whether persistence outcomes are correlated across many species after two assembled communities are mixed. We added this conceptual bridge to Supplementary Note 7, together with an auxiliary gLV analysis showing that excess same-parent shared fate increases with the interaction-strength parameter μ (Pearson $r = 0.870$, $p = 3.2 \times 10^{-8}$; Supplementary Fig. 36).

Finally, we keep the metric-specific interpretation in Results §2.2 and Supplementary Note 7, while using the Discussion to state the broader scope of the term:

“Accordingly, we use community-level selection as an outcome-level description of origin-correlated persistence during coalescence. This operational definition does not by itself identify the causal ecological or biochemical mechanisms, which may vary across systems.”

R2-5. Pre-selection of natural communities

The inclusion of natural communities is an important strength of the study. However, the use of a common stabilisation phase in defined media may introduce pre-selection effects that reduce ecological heterogeneity across communities.

In particular, incubating diverse natural communities in simplified media is known to drive convergence toward a limited set of functional guilds (Goldford et al., 2018). As a result, the “natural” communities used here may already be filtered to resemble the synthetic communities, potentially contributing to the similarity of results across systems.

We therefore suggest that the authors:

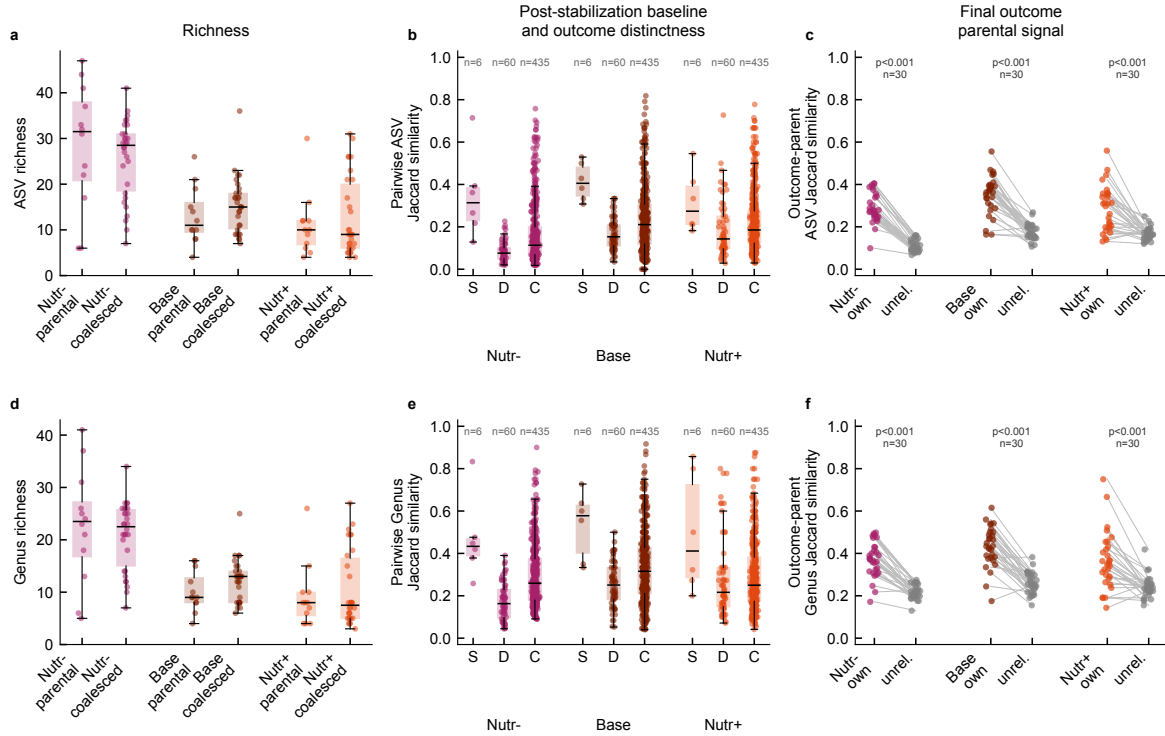
- discuss the potential effects of this pre-selection more explicitly
- clarify the extent to which taxonomic or functional convergence occurs during the stabilisation phase

This would help contextualise the generality of the natural-community results.

We agree this is an important concern. As the reviewer notes, defined-media stabilization can drive taxonomic or functional convergence (Goldford et al., 2018), and we cannot directly quantify pre-to-post taxonomic convergence or functional convergence here because the present experiments collected post-stabilization 16S community profiles rather than pre-enrichment, metagenomic, metabolomic, or trait-resolved measurements. We therefore narrow our claim: the natural sample-derived experiment shows that the nutrient-dependent increase in Dominance also occurs in taxonomically richer, laboratory-stabilized communities derived from natural samples, not that unmanipulated natural ecosystems follow the same rules.

Within the available data, we asked whether stabilized communities still carried source-specific taxonomic signal. Using same-source biological replicates as a comparison baseline, different-source parental communities remained substantially less similar at the ASV level (Jaccard similarity 1.7- to 4.1-fold lower than same-source replicates across the three media; Response Fig. R2-5b; Supplementary Fig. 22b), with a comparable pattern at the genus level (Response Fig. R2-5e; Supplementary Fig. 22e). This does not rule out functional convergence, or taxonomic convergence within each source relative to its original inoculum; only a pre-enrichment baseline could test those possibilities directly, and we do not have one. The data indicate that stabilization did not produce complete ASV- or genus-level taxonomic collapse to a single shared endpoint. Coalesced communities also retained parent-specific taxonomic signal, remaining closer to their own parents than to unrelated same-medium parents (all ASV- and genus-level tests $p < 10^{-6}$; Response Fig. R2-5c,f; Supplementary Fig. 22c,f).

We therefore limit the claim to nutrient-dependent Dominance in taxonomically richer laboratory-stabilized communities derived from natural samples, while noting that taxonomic or functional pre-selection during the seven-cycle stabilization may affect generalization to unmanipulated natural ecosystems. This revision also keeps the natural sample-derived community claims appropriately cautious.



Response Fig. R2-5. Natural sample-derived communities retain ASV- and genus-level distinctness after stabilization and coalescence. a,d, ASV- and genus-level richness by medium. b,e, Same-source stabilized parental replicates are more similar than different-source parental communities, indicating retained source-specific taxonomic signal after stabilization. c,f, Coalesced communities remain more similar to their own parental communities than to unrelated same-medium parental communities. Analyses use ASVs above the 0.1% relative-abundance threshold and post-stabilization 16S profiles; they test for complete post-stabilization taxonomic collapse and retained parent-specific signal, but not pre-to-post enrichment convergence or functional convergence.

We revised the manuscript to frame this experiment as applying to natural sample-derived communities established through laboratory stabilization, rather than to unfiltered natural ecosystems. We also added the same-source/different-source taxonomic-distinctness analysis to the Supplementary Information as Supplementary Fig. 22, and revised Results §2.6 to present pre-selection during stabilization as an alternative explanation for similarity between synthetic and natural sample-derived outcomes:

“These communities showed higher Restructuring fractions, possibly reflecting greater taxonomic diversity and more complex interaction networks. Because these communities were laboratory-stabilized before coalescence, pre-selection in defined media could have contributed to their similarity to synthetic-community outcomes. However, their higher richness and retained source-specific taxonomic structure indicate that stabilization did not produce complete ASV- or genus-level taxonomic collapse (Supplementary Figs. 19–22 and Supplementary Note 6). Overall, these results suggest that the nutrient-dependent increase in Dominance observed in synthetic consortia is also present in taxonomically richer natural sample-derived communities.”

We also added Supplementary Note 6 to state explicitly that pre-selection may occur and that the current data cannot quantify taxonomic convergence during stabilization or functional convergence.

R2-6. Frame gLV as phenomenological

The gLV model successfully reproduces the qualitative transition in coalescence outcomes and serves as a useful minimal framework. However, it is limited to purely competitive interactions and does not explicitly capture environmentally mediated effects such as pH modification.

We therefore suggest framing the model more explicitly as a phenomenological rather than mechanistic description of the system.

We agree that the gLV model should be framed as a phenomenological competitive baseline rather than as a mechanistic description of the biochemical interactions operating in the experiments. Results §2.2 (“Theoretical model with random interactions reproduces community-level selection”) now states:

“The gLV model serves as a phenomenological framework to explore how the statistical properties of interaction coefficients shape coalescence outcomes, rather than as a mechanistic model of the specific biochemical interactions (e.g., pH modification, metabolic cross-feeding) underlying our experimental observations.”

In addition, while the robustness analysis across interaction coefficient distributions is appreciated, a brief discussion of the biological interpretation of these distributions would be helpful. In particular, clarifying how the mean interaction strength parameter (μ) relates to underlying ecological processes would improve interpretability.

In the model, μ parameterizes the sampling range for competitive interaction coefficients in the gLV model. Larger μ makes the competitive coefficient ensemble stronger and more heterogeneous, moving the system away from a near-no-interaction regime and creating stronger mismatches between species that assembled together and outsiders. This lets assembly filter species into mutually compatible groups, the “cohesion without cooperation” route to Dominance discussed in the revised Discussion and Supplementary Note 3.

We decoupled these scale and heterogeneity effects in Supplementary Note 3 and Supplementary Fig. 42. This sweep shows that both average inhibitory effect and coefficient heterogeneity contribute to the Dominance regime. This is consistent with the baseline random-interaction model, where the uniform distribution $U(0, 2\mu)$ couples coefficient mean and spread.

We also revised the Results, Methods, and Supplementary Methods to avoid presenting μ as simply a mean interaction strength and to define it instead as the sampling-range parameter for competitive interaction coefficients. Results §2.2 now states:

“We fixed $r_i = 1$ and drew off-diagonal competition coefficients from a uniform distribution $\mathbb{U}(0, 2\mu)$, so increasing μ expands the sampled coefficient range and increases both the mean and spread of competitive effects (Hu et al., 2022, 2025).”

Methods now states:

“We fixed growth rates to 1 and self-interaction coefficients $\alpha_{ii} = 1$, and drew off-diagonal competition coefficients from a uniform distribution $\mathbb{U}(0, 2\mu)$, so μ sets the sampled coefficient range and

therefore controls both the mean and heterogeneity of competitive effects within this phenomenological model.”

Supplementary Methods further defines α_{ij} as an effective per-capita inhibitory term and states that the baseline $\alpha_{ij} \geq 0$ model captures competitive growth inhibition rather than facilitation.

R2 minor comments

Improve the visualisation of pairwise correlation results, as the separation between groups is not visually clear.

We agree that the previous visualization could be clearer. We revised the Fig. 2D presentation and caption to make the group summaries and null baseline easier to interpret:

“Individual dots show per-event pairwise selection correlations; in the simulation panel, 100 stratified events per group are displayed for legibility. Squares with error bars show mean $\pm$ s.e.m. across all coalescence events ($n = 1,200$ per group in simulations and all valid experimental events in the experimental panel). Gray horizontal lines indicate the random selection baseline from a null model in which parental-affiliation labels are shuffled (see Supplementary Information).”

Specify how pH was measured (e.g. continuously or at endpoints), and whether variability across replicates was assessed.

We clarified the pH-measurement description in Supplementary Methods, “Optical Density Measurements and Monoculture Characterization,” which now specifies the measurement timing:

“Community-level pH was measured from culture supernatants at the relevant endpoint, rather than continuously during growth.”

As with composition, OD, and outcome analyses, biological replicates were treated as separate event-level observations throughout.

Clarify the biological interpretation of the interaction coefficient distributions used in the theoretical model.

We clarified this point in Supplementary Methods, “Lotka–Volterra Simulations.” The revised text now explains that α_{ij} is an effective coefficient in the per-capita growth-rate equation that can absorb direct and indirect inhibitory effects, but does not identify a specific biochemical mechanism or dynamically model pH.

In the same Supplementary Methods section, we also clarified the meaning of μ and the distributional assumption:

“Thus, in the baseline competitive ensemble, μ sets the range of sampled competitive interaction coefficients. This parameter changes the overall scale and heterogeneity of pairwise coefficients, rather than isolating a single biological mechanism.”

Ensure consistent terminology and notation throughout (e.g. “pairwise”, “community-level”, α_{ij}).

We standardized the terminology throughout the revised manuscript. In particular, Results §2.2 now uses “pairwise selection correlation” for the species-pair fate-correlation metric and defines it as follows:

“We quantified this effect using pairwise selection correlation, which tests across coalescence events whether species pairs from the same parental community share survival outcomes (both persist or both go extinct) more strongly than species pairs from different parental communities (Supplementary Information; Fig. 2D).”

We use “community-level selection” for origin-correlated persistence at the parental-community level and use α_{ij} consistently for the gLV interaction coefficients.

Correct minor typographical issues, including “generalis ability” $\rightarrow$ “generalisability”.

We corrected the “generalis ability” typographical error and addressed other minor typographical issues throughout the revised manuscript.

Response to Reviewer 3

We thank the reviewer for their rigorous and technically insightful feedback, particularly the geometric analysis of dimensionality effects on similarity metrics. Below we address each point.

R3-1. Clarify L_1 versus L_2 normalization in the similarity metric

With respect to the similarity metrics used by the authors (the dot products $x_A \cdot x_C$ and $x_B \cdot x_C$), I believe that their interpretation is less straightforward than it would appear at first glance. As a first small comment, the authors state that x are “normalized abundance vectors,” and Fig. 1B clearly suggests that these are L_1 normalized, that is, their elements sum to 1 (i.e., they represent species’ relative abundances). But in the Supplementary Methods, the authors clarify that x are L_2 normalized, i.e., it is the sum of squares of their elements that is 1. I think making this more clear in the main text would help, since it fundamentally changes what one should expect for the dot products under different hypotheses.

We thank the reviewer for identifying this ambiguity and apologize for the confusion. To clarify, $\vec{x}$ denotes a community composition vector over ASVs, and the similarity between the coalesced community C and a parent P was computed as $\text{Sim}(C, P) = (\vec{x}_C \cdot \vec{x}_P) / (\|\vec{x}_C\|_2 \|\vec{x}_P\|_2)$. Because cosine similarity is invariant to multiplying either vector by a positive scalar, converting each sample from a raw ASV count vector to a relative-abundance vector before this step gives the same similarity score; the metric depends on the vector direction, not total abundance. We have corrected the Results, Methods, and Supplementary Methods (Coalescence Outcome Classification) wording accordingly. In the experimental analyses, these composition vectors are derived from the 16S ASV profiles.

Results §2.1, similarity-score definition:

“We represent each community by a community composition vector and compute the cosine similarity between the coalesced community C and each parental community (Fig. 1B):”

The revised equation explicitly divides each dot product by the product of the two Euclidean norms, and the following sentence defines $\vec{x}_A$, $\vec{x}_B$, and $\vec{x}_C$ as community composition vectors.

Methods §16S rRNA Sequencing now states:

“In experimental analyses, the community composition vector was the ASV relative-abundance vector derived from these 16S profiles.”

Methods §Classification of Coalescence Outcomes, at first definition of the similarity metric:

“Similarity was computed as cosine similarity between community composition vectors (see Supplementary Methods, Coalescence Outcome Classification, for full normalization details).”

Supplementary Methods, Coalescence Outcome Classification, metric definition paragraph:

“To characterize coalescence outcomes, we used cosine similarity as the underlying compositional similarity metric between the post-coalescence community and each parental community.”

This section then gives the same equation as the Results and clarifies that sample-wise total-abundance normalization does not change the cosine-similarity score.

R3-2. Dimensionality artifact in similarity metrics

Along these lines, my main concern is that these similarity metrics can behave in somewhat counter-intuitive ways, and some of the results the authors report may be at least partially skewed by simple statistical effects. To make my point as clear as possible, let me propose a simple example. Let's consider two communities A and B, each with N non-overlapping species. Total species' abundances can be represented by vectors n_A and n_B :

$$n_A = (n_A^{(1)}, n_A^{(2)}, \dots, n_A^{(N)}, 0, 0, \dots, 0)$$

$$n_B = (0, 0, \dots, 0, n_B^{(1)}, n_B^{(2)}, \dots, n_B^{(N)})$$

Now consider a simple null model (similar to the “shuffled abundance” null model proposed by the authors) in which community C, which results from A and B undergoing coalescence, has N species. The abundance vector of C, n_C , has N elements that are zero and N that are non-zero. Which elements are set to zero is chosen randomly (though the argument holds in other scenarios, including the “abundance-weighted random selection” model formulated by the authors). The non-zero elements of n_C are random samples of the non-zero elements of n_A and n_B . This null model aims at representing a “maximal restructuring” situation, where species' abundances in community C are completely uncorrelated from their abundances in the parental communities. If we generate n_A and n_B by randomly sampling their non-zero elements uniformly in $(0, 1)$ and we compute the similarity metrics

$$\text{Sim}(A, C) = n_A \cdot n_C / |n_A| |n_C|$$

$$\text{Sim}(B, C) = n_B \cdot n_C / |n_B| |n_C|$$

For different values of N and different pairs of randomly-generated communities, the “similarity map” looks like:

[See attached Reviewer Figure 1]

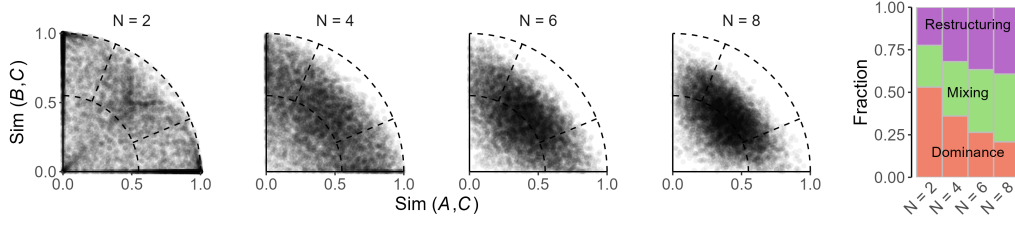
So despite the model being formulated as a null baseline representing a “full restructuring” situation, many cases would be flagged as “dominance” or “mixing.” The same is true if species' abundances in A and B are sampled differently, for instance if instead of sampling them uniformly ($n \sim U(0, 1)$) we used a more uneven distribution (such as $n \sim 10^{U(-3, 0)}$) we would find:

[See attached Reviewer Figure 2]

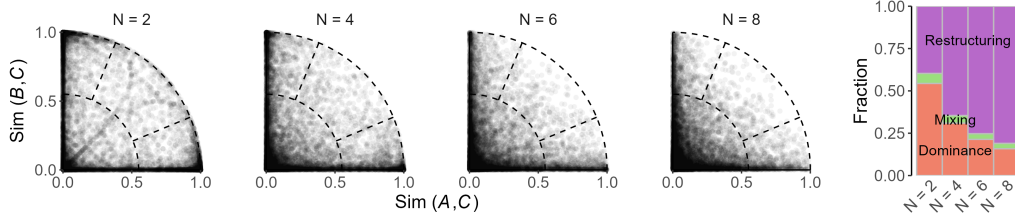
Notably, a somewhat similar effect also occurs under a much simpler null model: if the community C was simply the superposition of communities A and B (i.e., if the two parental communities mixed perfectly without interacting), we would have $n_C = n_A + n_B$. In this case:

[See attached Reviewer Figure 3]

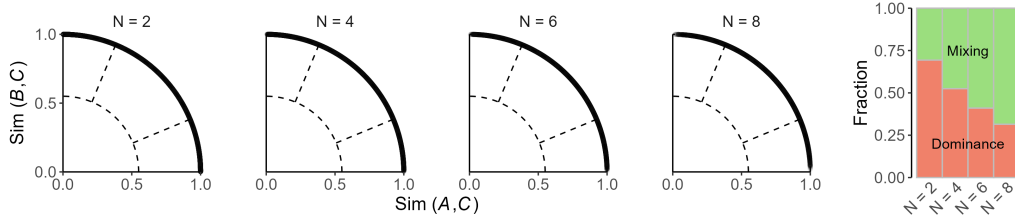
The key takeaway here is that, when using these similarity metrics and the author's “Dominance-Mixing-Restructuring” classification, one could expect to observe a decline in the “Dominance” fraction as community diversity increases due to a simple statistical effect: positive unit vectors are more likely to be close to orthogonal in lower dimensions than they are in higher dimensions.



Reviewer Figure 1



Reviewer Figure 2



Reviewer Figure 3

Reviewer attachment.

We thank the reviewer for raising this important point. We addressed this concern step by step. First, we reproduced the reviewer's toy null models with the manuscript classifier to identify the geometric and norm-imbalance effects that can inflate apparent Dominance classifications for purely statistical reasons. Second, we quantified the corresponding norm-imbalance effect in the actual experimental parental-community pairs and classified their simple additive null-model outcomes, showing that this effect is not sufficient to explain the observed Dominance tendency. Third, we performed the reviewer-requested case-by-case comparison between each observed coalescence outcome and its matched simple additive null, now included in the revised manuscript. We address the corresponding simulation concern below.

First, in the toy-null analysis, we reproduced the reviewer's toy null models using the same L_2 /cosine-similarity and vector-decomposition classification pipeline used in our manuscript (Response Fig. R3-2A). We also added the corresponding skewed-abundance additive case (random restructuring with $U(0,1)$, random restructuring with $10^{U(-3,0)}$, simple additive mixing $n_C = n_A + n_B$ with abundances drawn from $U(0,1)$, and simple additive mixing $n_C = n_A + n_B$ with abundances drawn from $10^{U(-3,0)}$). We note that one difference from the reviewer's attached

visualization is that the Restructuring boundary appears to use a retention-radius cutoff of $r < 1/2$, whereas our pipeline uses $r \leq 1/\sqrt{2}$ (equivalently, $r^2 \leq 1/2$).

Using the same classification pipeline as in the manuscript, the reproduced results show that the reviewer’s concern applies most strongly to abundance-skewed null communities. The random-restructuring null models become increasingly classified as Restructuring as N grows (uniform case: 43.4%, 64.9%, 78.5%, 88.7% for $N = 2, 4, 6, 8$), and the uniform additive null model concentrates on the symmetric Mixture axis and is classified as Mixture in 80.5%, 97.4%, 99.1%, 99.7% of cases. In contrast, the skewed-abundance simple additive null often remains above the retention threshold, falls off the symmetry line, and can therefore be classified as Dominance (Dominance: 73.9%, 50.0%, 39.7%, 30.6% for $N = 2, 4, 6, 8$). This confirms that skewed abundance distributions can generate apparent Dominance under simple null models, motivating the coalescence-pair-specific simple additive null and norm-imbalance analyses below.

We have revised the Methods and Supplementary Methods to state this boundary explicitly, and Response Fig. R3-2A shows both the manuscript boundary and the alternative visual boundary at $r = 1/2$.

Methods, “Classification of Coalescence Outcomes,” now states that

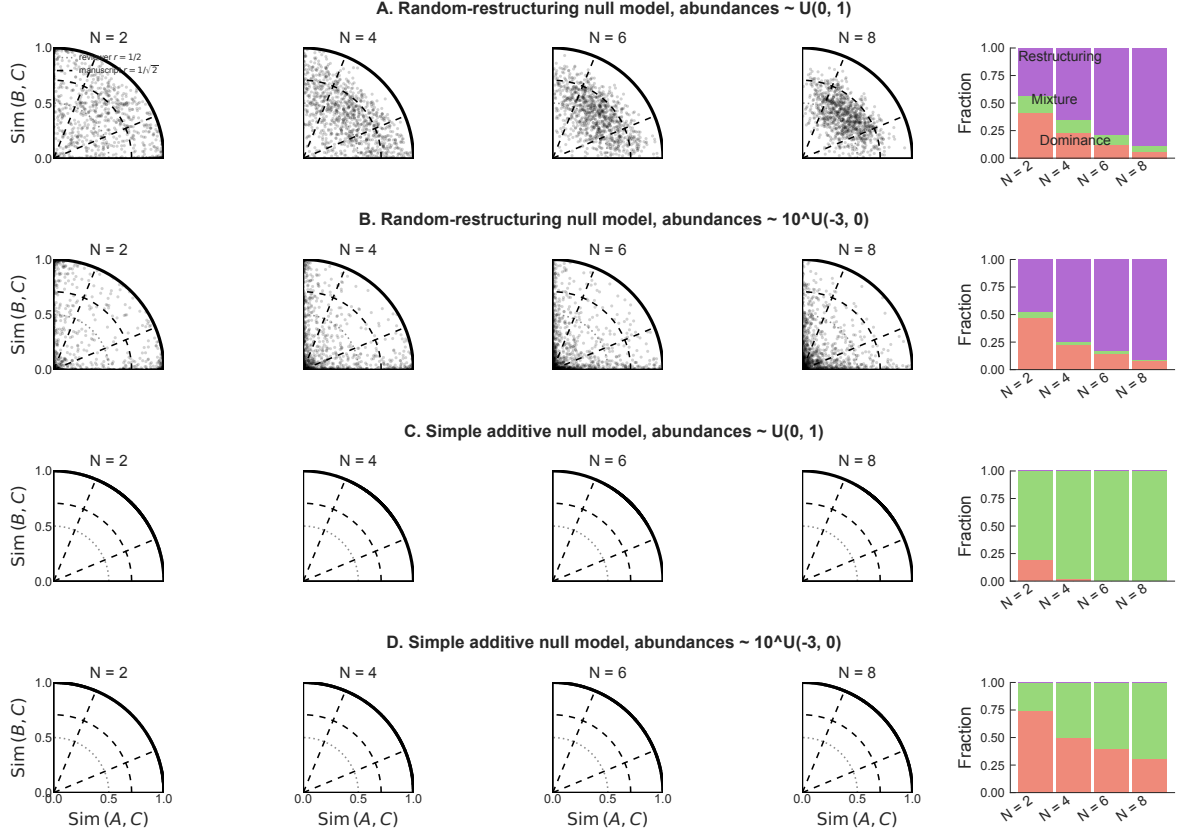
“The Restructuring boundary is defined by $r^2 \leq 0.5$, equivalently by a retention radius $r \leq 1/\sqrt{2}$ in the similarity map.”

Supplementary Methods, “Coalescence Outcome Classification,” now states that

“Restructuring occurs when $r^2 \leq 0.5$, equivalently when the retention radius satisfies $r \leq 1/\sqrt{2}$ in the two-parental-community similarity map, meaning that parental-community retention is low in the normalized similarity space, indicating substantial ecological reorganization.”

and

“First, the retention threshold $r^2 = 0.5$ (radius $r = 1/\sqrt{2}$, not $r = 1/2$) defines a coefficient-space retention boundary in the two-parental-community similarity map. For the special case of non-overlapping, orthonormal parental composition vectors, this boundary corresponds to half of the squared similarity-space mass lying along the two parental axes; with overlapping parental vectors, it should be interpreted as an operational retention-score boundary rather than a literal explained-composition fraction.”



Response Fig. R3-2A. Reviewer toy nulls analyzed with the manuscript classifier. Rows: **A**, random restructuring with $U(0, 1)$ abundances; **B**, random restructuring with $10^{U(-3, 0)}$ abundances; **C**, additive mixing with $U(0, 1)$ abundances; **D**, additive mixing with $10^{U(-3, 0)}$ abundances. Scatter panels show the reviewer’s visual boundary ($r = 1/2$, gray) and the manuscript boundaries (retention radius $r = 1/\sqrt{2}$ and asymmetry class boundaries, black); bars show manuscript-classifier fractions.

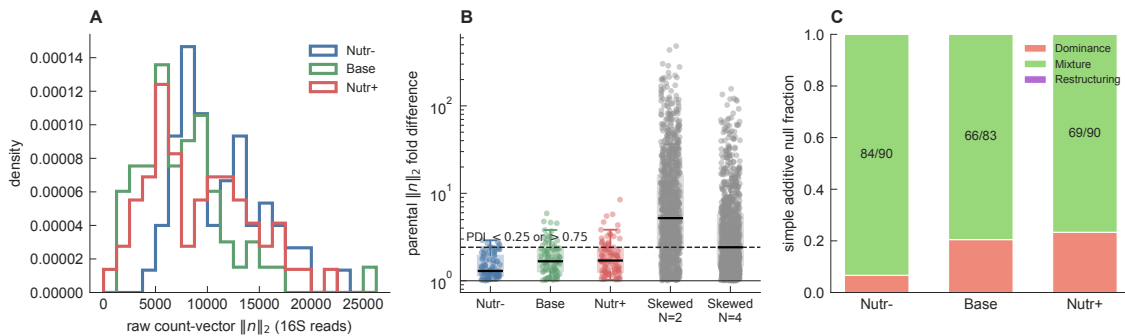
This effect is partially accounted for in the manuscript; specifically, Extended Data Fig. 3 shows that the empirical distribution of Dominance values has significantly higher mean than its counterparts generated by the null models. However, while the authors’ claim is correct (“skewed abundance distributions alone do not fully account for the observed asymmetric outcomes”), this does not mean that the empirically quantified Dominance values aren’t biased by the statistical effects described above. Many cases flagged as “Dominance” may in actuality be compatible with a simple null model, particularly for the communities assembled from the less diverse species pools.

The toy analysis shows that a simple additive null model with skewed abundance can be classified as Dominance because of a simple statistical effect. Specifically, this occurs when the two parental-community abundance vectors have substantially different norms. For a simple additive null model, $n_A + n_B$ is closer to the larger-norm parental community. For non-overlapping parental-community vectors, this effect is direct: $\text{Sim}(A, A+B)$ and $\text{Sim}(B, A+B)$ are proportional to $\|n_A\|_2$ and $\|n_B\|_2$, respectively. We agree that this effect could particularly matter when abundance is strongly uneven, because vector norms become sensitive to concentration of abundance among a small number of taxa.

However, the norm-imbalance effect is much weaker in our experimental parent pairs than in the

skewed toy examples. In our experimental coalescence pairs, the fold differences in Euclidean norms between the two parental-community raw ASV count vectors were 1.85 ± 0.95 (mean $\pm$ SD; median 1.58), with per-medium means ranging from 1.55 ± 0.51 to 2.05 ± 1.18 . These values are far smaller than those in the skewed-abundance simple additive null model, where the fold differences were 18.2 ± 39.4 and 7.1 ± 14.7 for $N = 2$ and $N = 4$, respectively (Response Fig. R3-2B).

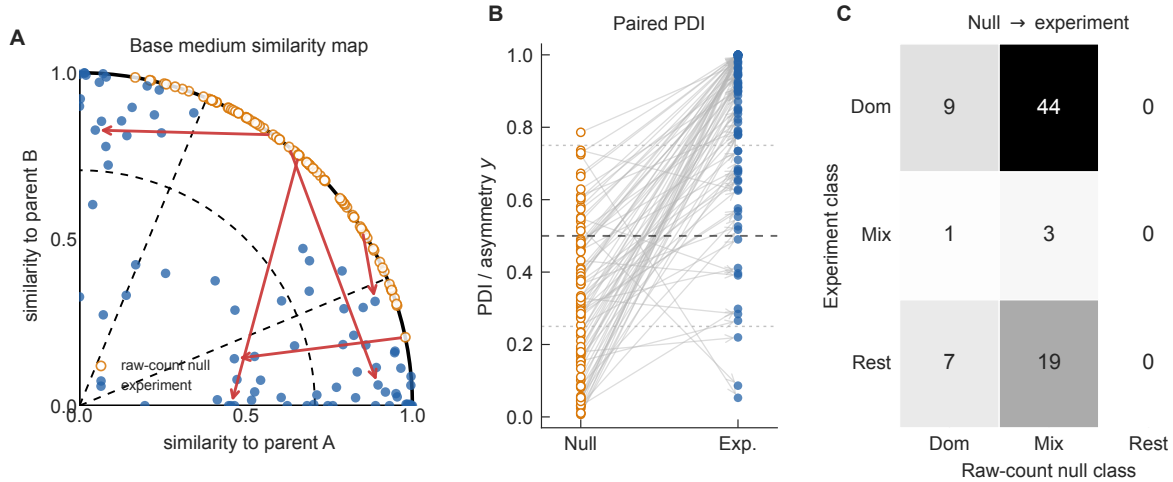
To directly test whether this effect could account for the experimental classifications, we constructed a simple additive null model from the raw ASV count vectors of the same two parental communities in each experimental coalescence pair (Response Fig. R3-2B). The simple additive null model was classified as Mixture in most events in each nutrient condition: 84/90 in Nutr− (93.3%), 66/83 in Base (79.5%), and 69/90 in Nutr+ (76.7%). Thus, the effect identified by the reviewer is present in principle and detectable in some experimental parent pairs, but it is not sufficient to explain the observed Dominance tendency.



Response Fig. R3-2B. Parental norm imbalance and simple additive null model classifications. **A**, L_2 norms of parental-community raw ASV count vectors. **B**, Paired parental L_2 norm fold differences, with skewed-abundance simple additive null models shown for scale; dashed line, $1 + \sqrt{2}$ fold difference corresponding to PDI < 0.25 or > 0.75 for the simple additive null model. **C**, Simple additive null model class fractions; labels show Mixture counts.

In my view, it would be necessary to compare each coalescence community with a corresponding null model constructed specifically for each of the pairs of parental communities, such as the simple additive model $n_{C,\text{null}} = n_A + n_B$. Comparing the experimental coalesced community with such a null model should be done on a case-by-case basis, not just by comparing distributions of Dominance indexes as in Extended Data Fig. 3.

We thank the reviewer for this suggestion. We compared each observed coalescence community with the corresponding simple additive null model, constructed from the same two parental communities. As described above, this null model was usually classified as Mixture, although it produced Dominance in a subset of events. Critically, the observed Base-medium outcomes were more strongly biased toward one parental community than their corresponding null-model outcomes (one-sided paired Wilcoxon test, $p = 5.9 \times 10^{-14}$). Thus, the coalescence-pair-specific comparison shows that the simple additive null model can contribute to apparent one-parent bias in some cases but does not by itself explain the observed Dominance tendency. We replaced Extended Data Fig. 3 with this coalescence-pair-specific simple additive null analysis and described the associated norm-imbalance effect in Supplementary Note 1.



Response Fig. R3-2C. Base-medium observations versus the coalescence-pair-specific simple additive null model. **A**, Similarity map: orange open points, simple additive null model outcomes; blue filled points, observed coalesced communities; arrows, example matched pairs. **B**, Paired asymmetry shifts from null to observation for all 83 Base events; gray lines connect matched null-observation pairs. **C**, Class transitions from simple additive null model outcomes (columns) to observations (rows).

R3-3. Interaction strength, diversity, and Dominance frequency

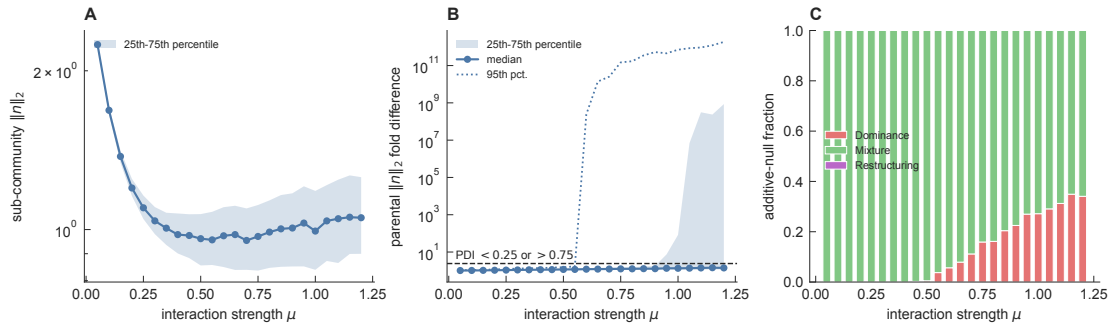
In section 2.3 of the Results, the authors examine how varying the mean interaction (competition) strength between species in a gLV model affects coalescence outcomes. The main conclusion here is that increasing competition strength shifts the distribution of coalescence outcomes from all “Mixing” to mostly “Dominance” and “Restructuring.” However, it is not clear to what extent this phenomenon can be explained by simple statistical effects like the ones described above. Increasing the mean interaction strength (μ) can be expected to decrease the number of species in the communities, and, as mentioned above, reducing N generically causes an increase in the frequency of “Dominance” outcomes. It seems important to test whether the increase of “Dominance” outcomes in the model (and in the experiments) as μ grows is substantially different from the expected increase in the null models as N decreases.

As shown in the R3-2 analyses, the richness- and norm-imbalance effect identified by the reviewer can occur for experimental parent-vector pairs under the additive null, but it is not sufficient to explain the nutrient-dependent experimental increase in Dominance. For the experimental data, the simple additive null model shows that paired parental raw ASV count vector norm differences were relatively small and that the simple additive null model was usually classified as Mixture rather than Dominance (Response Fig. R3-2B). Thus, this statistical effect contributes to the null expectation in some events but does not account for the experimental Dominance pattern.

In the gLV simulations, increasing interaction strength reduces final richness and creates greater unevenness in parental-community norms, so Dominance outcomes in the simple additive null model become more frequent because of this richness- and norm-imbalance effect (Response Fig. R3-3A). However, this effect is insufficient to explain Dominance in the gLV simulation coalescence outcomes. At the three representative interaction strengths used in the main figure, the simple additive null

model gives 0.0%, 5.7%, and 16.2% Dominance at $\mu = 0.3, 0.6$, and 0.8 , respectively, whereas the full gLV coalescence simulations give 20.4%, 61.0%, and 69.5% Dominance. Thus, richness- and norm-imbalance effects contribute to the null expectation at high μ , but they do not explain the simulated Dominance transition.

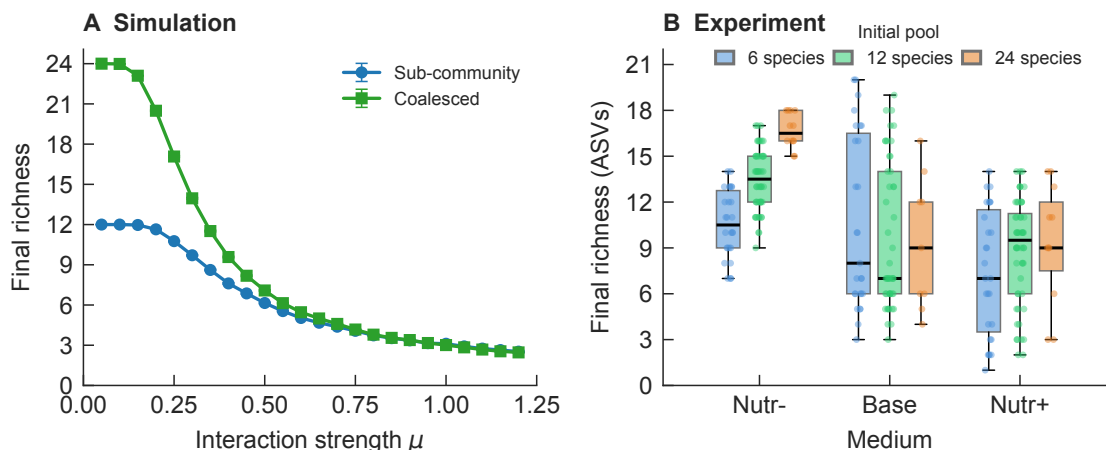
We have added this analysis to Supplementary Note 1, in the paragraph on geometric and parental-norm effects, and show the full analysis in Supplementary Fig. 38.



Response Fig. R3-3A. Simulation parental-community norm imbalance and simple additive null model classifications across interaction strength. **A**, Euclidean norms of assembled parental-community abundance vectors across μ ; line shows the median and shading shows the 25th–75th percentile range. **B**, Fold difference in Euclidean norm between the two parental communities for each simulated coalescence event; line shows the median and shading shows the 25th–75th percentile range, and dotted line shows the 95th percentile. The dashed horizontal line marks the fold-difference threshold corresponding to the Dominance boundary for the simple additive null model. **C**, Classification fractions for the simple additive null model computed from the same paired parental-community vectors at each μ .

It would also be useful to see how richness (in the final communities, not in the inocula) changes with μ in the model, as well as across the different media (Base, Nutr-, Nutr+) used in the experiment. From Supplementary Figs. 9-12, it seems like communities derived from natural samples do exhibit substantial diversity differences in the Base and Nutr+ media with respect to the Nutr- medium, but I could not find an equivalent representation for the first experiment.

We agree that showing final-community richness directly is useful. In the simulations, final richness decreases with μ for both post-assembly parental communities and coalesced communities (Response Fig. R3-3B, panel A). In the experiment, final coalesced-community richness also differs across media, with mean richness of 13.0, 9.9, and 8.5 ASVs in Nutr-, Base, and Nutr+, respectively (Response Fig. R3-3B, panel B). Thus, richness differences are present and are now shown directly. However, as demonstrated by the prior analyses, the associated statistical or geometric effect does not explain the Dominance patterns: it is limited in the experimental coalescence-pair-specific simple additive null analysis (Response Fig. R3-2B,C) and insufficient to explain the simulated Dominance transition (Response Fig. R3-3A).



Response Fig. R3-3B. Final richness across interaction strength and nutrient conditions. **A**, Mean final richness of simulated post-assembly parental communities and coalesced communities across μ ; error bars show s.e.m. **B**, Final coalesced-community richness (ASVs above 0.1% threshold) across nutrient conditions and initial pool sizes in the synthetic-community experiment.

R3-4. gLV model excludes facilitation

With respect to the model, the gLV framework is restricted to competitive interactions ($\alpha_{ij} > 0$). While this is reasonable from a modeling perspective because it prevents situations of “unchecked growth,” it also limits the range of ecological scenarios that the model can represent. Previous works have shown that facilitative interactions (particularly in the form of metabolic cross-feeding) are an important determinant of correlated species’ fates during coalescence.

We thank the reviewer for raising this important and valid limitation. We treat competition as the dominant interaction mode in this experimental system and in our modeling, while recognizing that facilitation can be important in other microbial coalescence settings. This approximation is supported by the 12-isolate pairwise coculture assay, where focal ASV yields were usually lower in coculture than in monoculture: 79/84/93 % of species-in-pair observations fell below the monoculture expectation in Nutr−/Base/Nutr+ (Response Fig. R3-4a).

We agree that the competition-only gLV model cannot capture facilitative interactions such as metabolic cross-feeding. To consider the broader ecological context, we ran the same coalescence analysis for two model extensions: (1) facilitative tails that allow positive ecological interactions (negative α_{ij} under our sign convention), and (2) reciprocal pair-coupling structure between coefficients α_{ij} and α_{ji} . The simulation definitions and full parameter sweeps are now described in Supplementary Note 3 and Supplementary Figs. 39–40.

In the facilitative-tail extension, we preserved the mean interaction coefficient and added density-dependent self-limitation to prevent runaway growth (Response Fig. R3-4b). Dominance still rises with μ at every facilitative-tail strength tested. At $\mu = 0.30$, increasing the facilitative tail shifts outcomes away from Mixture toward both Dominance and Restructuring (Dom/Mix/Res = 18/73/9 % at $f = 0$ and 43/33/24 % at $f = 0.8$). At $\mu = 0.80$, Dominance remains high across the facilitative tail (70–76%). Thus, facilitative interactions can change the balance between Mixture and Restruc-

turing in some regimes, but the central trend, increased Dominance with stronger interaction (μ), remains largely valid in this model extension.

We also varied reciprocal pair-coupling structure, using antisymmetric exploitation for $p < 0$ and symmetric competition for $p > 0$ (Response Fig. R3-4c; full fine sweep in Supplementary Fig. 40). These pair-coupling changes alter the detailed outcome fractions but do not overturn our core claim: Dominance still increases with stronger interaction (μ) under both antisymmetric exploitation and symmetric competition.

We added this model-extension result to Results §2.3:

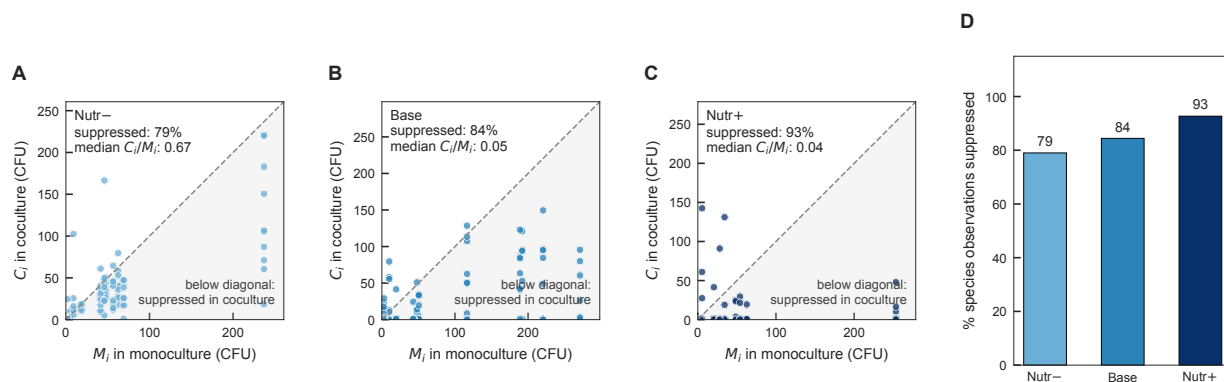
“These patterns were robust to variation in growth rates, carrying capacities, interaction-coefficient distributions, parental-community species richness, and model extensions allowing beneficial interactions or varying reciprocal pair-coupling structure (Supplementary Figs. 4–6 and 39–41; Extended Data Fig. 4).”

We clarified the baseline-model scope in the Supplementary Methods:

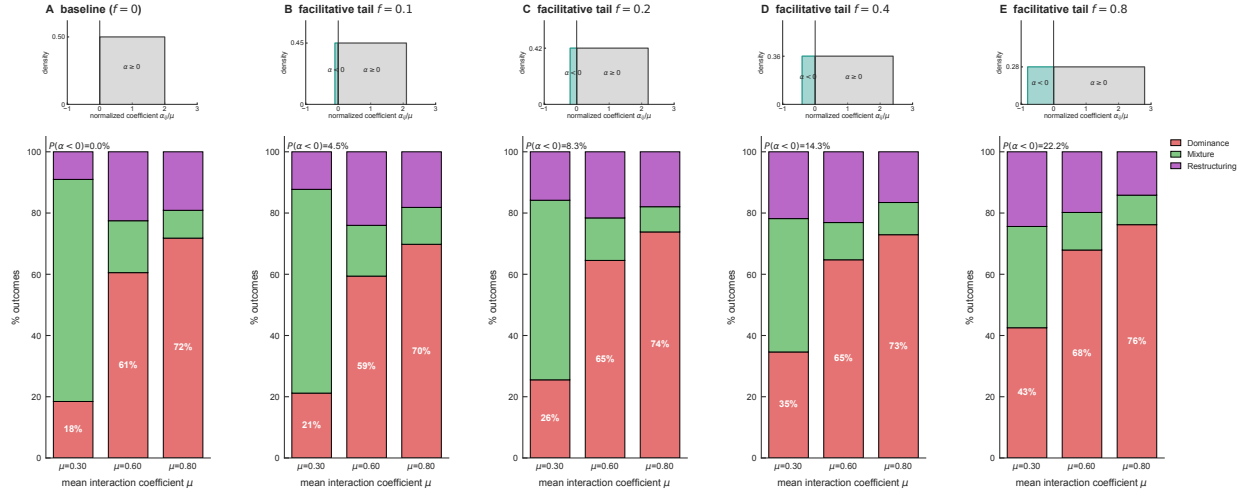
“Because the baseline model restricts $\alpha_{ij} \geq 0$, it captures only competitive (growth-inhibiting) interactions; facilitative interactions such as cross-feeding are not represented in the baseline ensemble.”

The Discussion now states the corresponding interpretation and future direction:

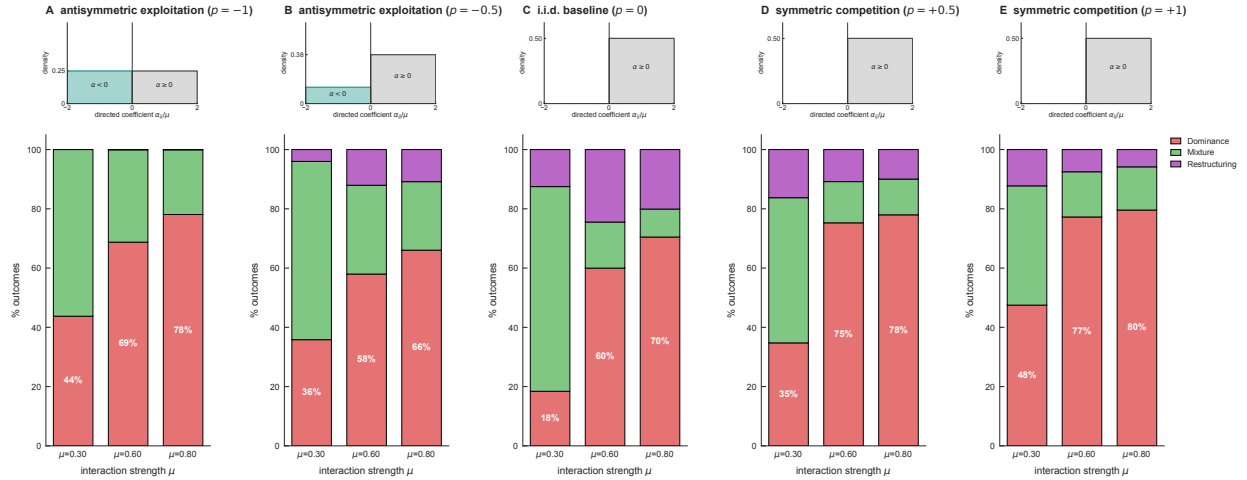
“Thus, cooperative and competitive mechanisms may represent distinct routes to community-level selection, and understanding how these routes interact remains an important direction for future work.”



Response Fig. R3-4a. Pairwise coculture data show widespread growth-inhibiting effects, especially in Base and Nutr+. (A–C) Coculture CFU (C_i) versus monoculture CFU (M_i) for each focal ASV in each unordered pair; dashed line, $C_i = M_i$. Points below the diagonal indicate lower yield in coculture. (D) Fraction of species observations below monoculture expectation: 79/84/93 % for Nutr–/Base/Nutr+.



Response Fig. R3-4b. Allowing a facilitative tail can increase Restructuring but does not remove the rise of Dominance with μ . Off-diagonal coefficients are sampled from $\alpha_{ij} \sim U[-f\mu, (2+f)\mu]$, preserving $E[\alpha_{ij}] = \mu$ while increasing the expected facilitative fraction $P(\alpha_{ij} < 0)$ from 0 to 22.2%. Top insets show labeled analytic coefficient distributions; stacked bars show Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$ (200 pools $\times$ 6 coalescence events per cell).



Response Fig. R3-4c. Dominance still increases with μ across reciprocal pair-coupling structures. Panels are ordered from antisymmetric exploitation ($p < 0$) through the iid competitive baseline ($p = 0$) to symmetric competition ($p > 0$). Top insets show labeled analytic directed-coefficient distributions; stacked bars show Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$ (200 pools $\times$ 6 coalescence events per cell).

Would two species which engage in a mutualism be considered “weak interactors” or the opposite?

Increasing the fraction or magnitude of mutualistic or facilitative coefficients can increase effective

interaction strength when it broadens the interaction-coefficient distribution, because coefficient variance is one component of our interaction-strength axis.

To test whether allowing positive ecological interactions affected the model result, we added weak bidirectional mutualistic pairs to the gLV simulations. Specifically, 0–40% of unordered species pairs were made bidirectionally mutualistic ($\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$), while the remaining interactions were left competitive. The μ -dependent Dominance trend persisted: Dominance remained 58.5–70.2% at $\mu = 0.60$ and 69.4–77.5% at $\mu = 0.80$ (Response Fig. R3-4e; Supplementary Fig. 41). Thus, adding weak mutualistic pairs does not remove the Dominance trend.

More generally, within our framework, interaction strength refers to coefficient magnitude and distributional spread/variance (the May-style random-Lotka–Volterra axis; May, 1972; Grilli et al., 2017), not interaction sign. Thus, mutualism is not intrinsically weak: two species in a strong mutualism would be strong interactors, with coefficients of opposite sign from strong competitors. We note that our empirical invasion-resistance assay, however, measures growth-inhibiting effects and would not score mutualistic facilitation as strong competitive exclusion.

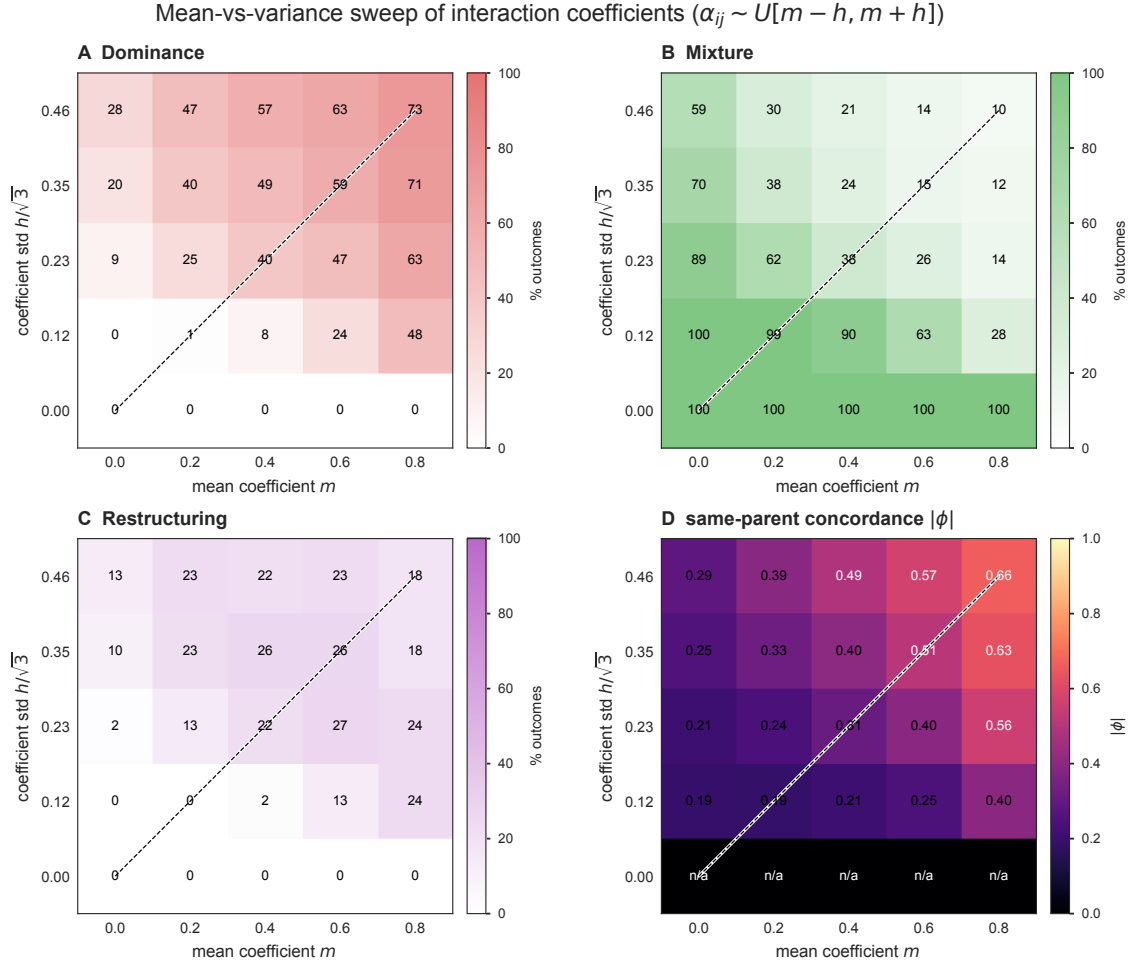
We have therefore revised the manuscript to make this definition clearer. Results §2.2 now states:

“We fixed $r_i = 1$ and drew off-diagonal competition coefficients from a uniform distribution $\mathbb{U}(0, 2\mu)$, so increasing μ expands the sampled coefficient range and increases both the mean and spread of competitive effects (Hu et al., 2022, 2025).”

Methods now states:

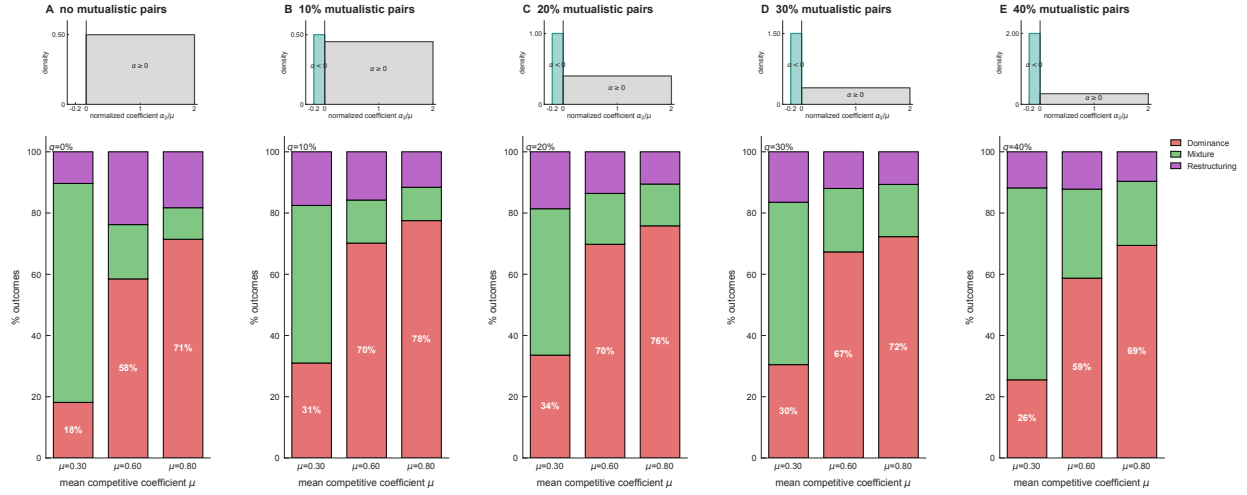
“We fixed growth rates to 1 and self-interaction coefficients $\alpha_{ii} = 1$, and drew off-diagonal competition coefficients from a uniform distribution $\mathbb{U}(0, 2\mu)$, so μ sets the sampled coefficient range and therefore controls both the mean and heterogeneity of competitive effects within this phenomenological model.”

In Supplementary Note 3, we further explain that changing this coefficient range affects both the average inhibitory effect and heterogeneity among species pairs, and analysis separating the mean and variance of the interaction-coefficient distribution shows that both components contribute to the Dominance regime (Response Fig. R3-4d; Supplementary Fig. 42).



Response Fig. R3-4d. Coefficient mean and variance jointly shape Dominance in the mixed-sign gLV sweep. Off-diagonal coefficients are sampled from $\alpha_{ij} \sim U[m - h, m + h]$, separating mean coefficient m from distributional spread h and therefore coefficient variance. The dashed line marks $h = m$, above which negative coefficients can occur. Heatmaps show Dominance, Mixture, Restructuring, and same-parent concordance $|\phi|$ across the 5×5 grid (200 pools $\times$ 6 coalescence events per cell).

Weak mutualistic-pair sweep ($\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$; $\gamma = 0.10$)



Response Fig. R3-4e. Weak bidirectional mutualistic pairs do not remove the μ -dependent Dominance trend. For a fraction q of unordered species pairs, both reciprocal coefficients are sampled from $\alpha_{ij}, \alpha_{ji} \sim U[-0.2\mu, 0]$; remaining coefficients are sampled from the baseline competitive distribution $U[0, 2\mu]$. Top insets show the labeled coefficient mixture; stacked bars show Dominance / Mixture / Restructuring fractions at $\mu \in \{0.30, 0.60, 0.80\}$ (200 pools $\times$ 6 coalescence events per cell).

This seems particularly relevant given that “Restructuring” appears to be much more common in natural sample-derived communities than in the initial bottom-up-assembled communities. Given these discrepancies, and the fact that natural sample-derived communities are still model systems (even if they may be closer to natural microbiomes than communities assembled from defined species pools), I would suggest the claims in section 2.6 of the Results as well as in the Discussion are toned down (e.g., lines 292-294).

We have changed and toned down our conclusion for natural sample-derived communities in Results §2.6:

“These communities showed higher Restructuring fractions, possibly reflecting greater taxonomic diversity and more complex interaction networks. Because these communities were laboratory-stabilized before coalescence, pre-selection in defined media could have contributed to their similarity to synthetic-community outcomes. However, their higher richness and retained source-specific taxonomic structure indicate that stabilization did not produce complete ASV- or genus-level taxonomic collapse (Supplementary Figs. 19–22 and Supplementary Note 6). Overall, these results suggest that the nutrient-dependent increase in Dominance observed in synthetic consortia is also present in taxonomically richer natural sample-derived communities.”

We also removed the detailed natural-community generality caveat from the Discussion and consolidated it in Supplementary Note 6, which emphasizes that these are laboratory-stabilized communities rather than unfiltered natural ecosystems:

“Thus, pre-selection could contribute to the convergent coalescence patterns observed between synthetic and natural sample-derived communities, and the natural-community results should be interpreted as evidence that the nutrient-dependent increase in Dominance occurs in taxonomically

richer laboratory-stabilized communities, not as evidence for unrestricted generality across unfiltered natural ecosystems.”

R3-5. “Interaction strength” versus “competition strength”

One last comment along these lines, which is admittedly mostly semantic: the term “interaction strength” may be somewhat confusing since, in the context of the gLV model, it refers strictly to competitive interactions. It may be worth considering using different terminology (“competition strength”?)

We agree that the term “interaction strength” can be ambiguous in our baseline gLV model, because the modeling framework uses non-negative, growth-inhibiting coefficients. We nevertheless retain “interaction strength” as the umbrella term because it is standard in the random-interaction and microbial gLV literature that motivates our framework, including May’s classic random community model (May, 1972) and recent microbial gLV work (Hu et al., 2022, 2025).

Instead, we revised the manuscript to define the baseline model more precisely. We now refer to μ as controlling the sampled range, mean, and heterogeneity of competitive interaction coefficients, and we explicitly state that the baseline gLV model is competition-dominated rather than a general model of all microbial interactions. Methods now states:

“We fixed growth rates to 1 and self-interaction coefficients $\alpha_{ii} = 1$, and drew off-diagonal competition coefficients from a uniform distribution $\mathbb{U}(0, 2\mu)$, so μ sets the sampled coefficient range and therefore controls both the mean and heterogeneity of competitive effects within this phenomenological model.”

As described in R3-4, we also added facilitative-tail, reciprocal pair-coupling, weak-mutualistic-pair, and coefficient mean/variance robustness analyses to define the scope of this terminology.

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